MEDIACREST TRAINING COLLEGE

Scholarship Program

AI and Machine Learning Foundation

Comprehensive Study Notes

Week 1, Day 1

Demystifying AI & Local Applications

Course Level:	Certificate (Skills-Based)
Duration:	6 Weeks
Mode:	Online and Physical
Intake:	March 2026

Prepared by:

Mediacrest Training College Faculty

AI & Machine Learning Department

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Nairobi, Kenya

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This material is exclusively for students enrolled in the AI and Machine Learning Foundation Program.

Contact Information:

Mediacrest Training College
Nairobi, Kenya
Email: info@mediacrest.ac.ke
Website: www.mediacrest.ac.ke

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Preface

Welcome to the **AI and Machine Learning Foundation Program** at Mediacrest Training College. This comprehensive study guide has been meticulously designed to accompany your learning journey through the exciting world of Artificial Intelligence.

Purpose of This Document

This 60-page study companion serves multiple purposes:

- **Foundation Building:** Provides clear, jargon-free explanations of complex AI concepts
- **Local Context:** Emphasizes AI applications specifically relevant to Kenya and East Africa
- **Practical Focus:** Bridges theory with real-world applications you can implement immediately
- **Reference Guide:** Serves as a lasting resource you can return to throughout your career
- **Self-Paced Learning:** Enables you to study at your own pace, reinforcing classroom learning

How to Use This Document

We recommend the following approach:

1. **Before Class:** Skim the relevant chapter to familiarize yourself with key concepts
2. **During Class:** Take notes directly in the margins and highlight important points
3. **After Class:** Review the chapter thoroughly and complete all practical exercises
4. **Before Assignments:** Use the glossary and examples to reinforce your understanding
5. **Ongoing Reference:** Keep this document as a professional resource for future projects

Document Structure

This document is organized into six comprehensive chapters:

Chapter 1: Course Overview and Orientation - Setting expectations and technical requirements

Chapter 2: The AI Family Tree - Understanding AI, ML, and DL relationships

Chapter 3: AI in the Kenyan Context - Local applications and case studies

Chapter 4: Navigating the AI Tool Landscape - Comprehensive tool guide

Chapter 5: The Art of Prompt Engineering - Mastering AI communication

Chapter 6: Applied Problem Solving - Framework for AI solutions

Appendices: Glossary, references, and assignment templates

A Note on Accessibility

This program is designed for learners with **little to no programming experience**. We have intentionally avoided heavy mathematics and complex technical jargon, focusing instead on conceptual understanding and practical application. Every effort has been made to ensure that the content is accessible, engaging, and immediately applicable to your personal and professional contexts.

The Mediacrest Training College Faculty
AI & Machine Learning Department
March 2026

Chapter 1

Course Overview and Orientation

Learning Objective

Chapter Learning Objectives:

- Understand the course structure, duration, and delivery mode
- Identify if you meet the entry requirements
- Set up your technical environment correctly
- Understand assessment methods and certification requirements
- Recognize career outcomes and opportunities

1.1 Course Description

The **AI and Machine Learning Foundation Program** is a comprehensive, practical, and hands-on introduction to Artificial Intelligence, specifically designed for individuals with little or no programming experience. This course represents a unique opportunity to gain cutting-edge skills that are transforming industries worldwide, with particular emphasis on applications relevant to the Kenyan and African context.

1.1.1 What Makes This Course Different?

Unlike traditional AI courses that focus heavily on theoretical mathematics and complex programming, our program emphasizes:

- **Practical Application:** Learn by doing, with hands-on projects that solve real problems
- **No-Code/Low-Code Approach:** Use powerful AI tools through intuitive interfaces
- **Local Relevance:** Case studies and examples from Kenyan businesses and communities

- **Business Focus:** Understand how AI can optimize operations, reduce costs, and drive innovation
- **Ethical Awareness:** Develop critical thinking about AI bias, fairness, and responsible use

1.1.2 Course Philosophy

We believe that AI literacy is no longer a luxury—it is an essential skill for the 21st century. Whether you are a student, business owner, marketing professional, or customer service representative, understanding AI will:

1. Enhance your productivity and decision-making capabilities
2. Open new career opportunities and pathways
3. Enable you to automate repetitive tasks and focus on high-value work
4. Help you understand and navigate an increasingly AI-driven world
5. Empower you to contribute to Kenya's technological advancement

1.2 Target Audience

This course has been specifically designed for five primary learner profiles:

1.2.1 Students and Career Switchers

If you are:

- A recent graduate looking to enter the tech industry
- A professional considering a career change into AI or data science
- Someone curious about technology trends and future skills

This course provides the foundational knowledge and practical skills to begin your AI journey without requiring a computer science background.

1.2.2 Business Owners and Entrepreneurs

As a business owner in Kenya's dynamic market, you will learn to:

- Identify opportunities for AI automation in your operations
- Reduce operational costs through intelligent systems
- Improve customer service with AI-powered tools
- Make data-driven decisions with confidence
- Stay competitive in an increasingly digital marketplace

1.2.3 Marketing, HR, and Administration Professionals

These departments are being transformed by AI. You will discover how to:

- Automate routine administrative tasks
- Analyze customer data for better marketing insights
- Streamline recruitment and HR processes
- Create compelling content with AI assistance
- Improve organizational efficiency

1.2.4 Customer Service Representatives

Customer service is being revolutionized by AI. Learn to:

- Work alongside AI chatbots and virtual assistants
- Use AI tools to respond faster and more accurately
- Analyze customer sentiment and feedback
- Personalize customer interactions at scale
- Enhance customer satisfaction and loyalty

1.2.5 General Interest Learners

Even if you don't fit the above categories, if you are curious about AI and want to understand:

- How AI impacts your daily life
- The opportunities and risks of AI technology
- How to use AI tools effectively and ethically

This course provides a comprehensive, accessible introduction.

1.3 Entry Requirements

To ensure your success in this program, we have established the following entry requirements:

1.3.1 Educational Background

- **Minimum Requirement:** High school completion (KCSE or equivalent)
- **Recommended:** Any diploma or degree (any field)
- **No Programming Experience Required:** This course is designed for beginners

1.3.2 Technical Requirements

Important

Critical: Ensure your computer meets these specifications **before** the course begins. Technical issues can significantly impact your learning experience.

Component	Requirement	Status
Operating System	Windows 10+, macOS 10.15+, or Linux	Verified
Storage	256 GB SSD minimum	Verified
RAM	4 GB minimum (8 GB recommended)	Verified
Internet	Stable broadband (5 Mbps minimum)	Verified
Browser	Latest Chrome, Edge, or Firefox	Verified
Webcam & Microphone	Required for interactive sessions	Verified

Table 1.1: Technical Requirements Checklist

1.3.3 Required Software and Accounts

All primary tools used in this course are **free and browser-based**:

1. **Google Account** (required for):
   - Google Colab (cloud-based Python environment)
   - Google Sheets/Excel Online (data handling)
   - Teachable Machine (no-code ML tool)
   - Google Drive (file storage and sharing)
2. **Generative AI Access** (choose one):
   - ChatGPT (free tier at chat.openai.com)
   - Google Gemini (free at gemini.google.com)
   - Microsoft Copilot (free at copilot.microsoft.com)
3. **Optional but Recommended**:
   - GitHub account (for code sharing and collaboration)
   - Kaggle account (for datasets and learning resources)
   - Hugging Face account (for AI models and tools)

1.4 Course Structure

The 6-week program is carefully structured to build your knowledge progressively:

1.4.1 Week-by-Week Breakdown

Week	Topic	Key Outcomes
1	AI Foundations & Literacy	Understand AI/ML/DL, recognize use cases
2	Data Basics	Clean and prepare datasets
3	ML Principles	Understand supervised vs unsupervised learning
4	Beginner ML Project	Build your first predictive model
5	AI for Work & Business	Apply AI to professional contexts
6	Capstone Project	Demonstrate comprehensive AI skills

Table 1.2: Course Schedule Overview

1.4.2 Detailed Weekly Content

Week 1: AI Foundations & AI Literacy

Topics Covered:

- What is AI, ML, and DL (simple analogies)
- Real-life AI applications in Kenya
  - Mobile lending algorithms
  - Chatbots in customer service
  - AI in agriculture
  - Fraud detection systems
- AI ethics and bias
- Overview of AI tools (ChatGPT, Gemini, Copilot, Teachable Machine, AutoML)

Practical Activities:

- Live demonstration: Using ChatGPT to solve a simple problem
- Assignment: Identify three AI tools you interact with weekly

Week 2: Data Basics

Topics Covered:

- What is data? Structured vs unstructured data
- Types of data (text, numbers, images)
- Introduction to datasets (CSV files)
- Data cleaning basics using Excel/Google Sheets
  - Removing duplicates
  - Handling missing values
  - Sorting and filtering
  - Data validation

Practical Activities:

- Clean a sample dataset and document the steps taken
- Create data quality reports

Week 3: Principles of Machine Learning

Topics Covered:

- Supervised vs Unsupervised Learning
- Classification vs Regression
- Training data vs Testing data
- How ML models "learn" conceptually
- Model evaluation metrics
- Overfitting and underfitting

Practical Activities:

- Interactive demonstrations of ML concepts
- Compare different learning approaches

Week 4: Beginner ML Project

Project Options (choose one):

- Predict student performance
- Predict customer churn
- Simple sales prediction

Tools:

- Google AutoML
- Microsoft Azure ML Studio
- Teachable Machine
- Optional: Light Python notebook track

Practical Activities:

- Step-by-step project walkthrough
- Upload dataset → train → evaluate → interpret results

Week 5: AI for Work and Business

Topics Covered:

- AI applications in:
  - Digital marketing
  - Customer support
  - Human resources
  - Content creation
  - Data analysis
- Basics of prompt engineering
  - Role definition
  - Context setting
  - Iterative prompting
- Introduction to AI automations (Zapier, Make.com)

Practical Activities:

- Create a simple AI-powered workflow
- Content generation task

Week 6: Capstone Project

Project Options (choose one):

- Build an AI-powered workflow or automation
- Train a simple ML model for a practical problem
- Develop a real-world AI solution (e.g., predicting sales, student performance, customer behavior)

Project Development Steps:

1. Define the problem or goal

2. Prepare or clean the dataset
3. Train and evaluate the model
4. Test and interpret results
5. Present findings or solution to the group

1.5 Assessment Methods

Assessment is conducted through continuous practical activities and formal assignments:

1.5.1 Weekly Assignments (30%)

Each week includes practical tasks designed to reinforce learning:

- Week 1: Identify AI tools in daily use
- Week 2: Document data cleaning processes
- Week 3: Compare ML approaches
- Week 4: Complete guided ML project
- Week 5: Create AI workflow

1.5.2 Guided Projects (30%)

Execute ML projects with specific outcomes:

- Sales prediction model
- Performance prediction system
- Customer behavior analysis

1.5.3 Capstone Project (40%)

The primary assessment involves:

- Developing a functional AI/ML mini-project
- Presenting the solution to the group
- Demonstrating practical application of concepts
- Written documentation of the project

1.6 Certification

Upon successful completion of the course and delivery of a functional Capstone Project, participants will receive:

- **Certificate of Completion** recognizing ability to apply AI and ML concepts
- **Digital Badge** for LinkedIn and professional profiles
- **Project Portfolio** showcasing your work
- **Letter of Recommendation** (for outstanding performers)

1.6.1 Certification Requirements

To qualify for certification, you must:

1. Attend at least 80% of sessions
2. Complete all weekly assignments
3. Submit a functional capstone project
4. Achieve a minimum overall score of 60%
5. Participate in group discussions and activities

1.7 Career Outcomes

Graduates of this program will be equipped to:

1.7.1 Enhance Productivity

- Leverage AI tools for automation
- Make better data-driven decisions
- Streamline workflows and processes
- Reduce time spent on repetitive tasks

1.7.2 Optimize Business Operations

- Apply AI in marketing campaigns
- Improve HR recruitment processes
- Enhance customer service delivery
- Optimize inventory and supply chain

1.7.3 Drive Innovation

- Design AI-powered solutions
- Identify new business opportunities
- Implement automation strategies
- Lead digital transformation initiatives

1.7.4 Potential Career Paths

Role	How This Course Helps
AI Consultant	Understand AI capabilities and limitations
Business Analyst	Apply AI to solve business problems
Marketing Specialist	Use AI for campaign optimization
Customer Success Manager	Implement AI chatbots and automation
Data Analyst	Clean, analyze, and interpret data
Product Manager	Design AI-powered features
Entrepreneur	Build AI-enabled products/services

Table 1.3: Career Pathways

1.8 Scholarship Information

1.8.1 Scholarship Coverage

This scholarship program is fully sponsored by our development partner:

- **Tuition Fees:** Fully covered
- **Learning Materials:** Provided at no cost
- **Software Access:** Free tools and platforms

1.8.2 Statutory Charges

However, to facilitate academic requirements, the following statutory charges apply:

Item	Amount (KES)
Registration	1,000
1st Installment (27th March)	3,500
2nd Installment (13th April)	3,500
Total	**8,000**

Table 1.4: Payment Schedule

1.8.3 Payment Details

Example

Account Information:
Account Name: Mediacrest Training College
Account Number: 236692
Bank: [Bank Name]
Branch: [Branch Name]
Reference: Your Full Name + AI/ML March 2026

1.9 Student Support Services

1.9.1 Academic Support

- **Office Hours:** Faculty available for one-on-one consultations
- **Peer Learning:** Study groups and collaborative projects
- **Online Forum:** 24/7 discussion platform
- **Resource Library:** Access to additional learning materials

1.9.2 Technical Support

- IT helpdesk for technical issues
- Software installation assistance
- Platform troubleshooting
- Backup and recovery support

1.9.3 Career Support

- Resume and portfolio review
- Interview preparation
- Job placement assistance

- Industry networking events

1.10 Course Policies

1.10.1 Attendance Policy

- Minimum 80% attendance required
- Notify instructor in advance of absences
- Make-up sessions available for excused absences
- Online sessions recorded for review

1.10.2 Assignment Submission

- All assignments must be submitted on time
- Late submissions accepted with 10% penalty per day
- Maximum 3 days late acceptance
- Submit through designated platform only

1.10.3 Academic Integrity

- Plagiarism strictly prohibited
- Cite all sources and references
- Collaborate ethically on group work
- Violations result in disciplinary action

1.10.4 Code of Conduct

- Respect for all participants
- Professional communication
- Inclusive and supportive environment
- Confidentiality of shared information

Chapter 2

The AI Family Tree: Understanding Core Concepts

Learning Objective

Chapter Learning Objectives:

- Define Artificial Intelligence (AI) clearly and accurately
- Explain Machine Learning (ML) and its three main types
- Understand Deep Learning (DL) and neural networks
- Differentiate between AI, ML, and DL
- Contrast traditional programming with machine learning
- Apply the "Fruit Sorting" analogy to real problems

2.1 Introduction to Artificial Intelligence

2.1.1 Historical Context

Artificial Intelligence is not a new concept. The journey began in the 1950s when pioneers like Alan Turing asked the fundamental question: "Can machines think?" Since then, AI has evolved through several phases:

- **1950s-1970s:** Early optimism and symbolic AI
- **1980s-1990s:** Expert systems and knowledge-based approaches
- **2000s-2010s:** Machine learning and data-driven methods
- **2010s-Present:** Deep learning and neural networks

2.1.2 What is Artificial Intelligence?

Formal Definition: Artificial Intelligence is the simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, problem-solving, perception, and language understanding.

Core Capabilities of AI

AI systems exhibit several key capabilities that mimic human intelligence:

1. **Learning:** The ability to acquire information and formulate rules for converting that information into actionable knowledge.
    - Example: A spam filter learning which emails are junk based on your past behavior
    - Example: Netflix learning your viewing preferences over time
2. **Reasoning:** The ability to use established rules to reach approximate or definite conclusions.
    - Example: A chess AI calculating the best move based on game rules
    - Example: A medical diagnosis system suggesting possible conditions
3. **Problem-Solving:** The ability to identify a goal and determine a sequence of actions to achieve it.
    - Example: GPS finding the optimal route from point A to point B
    - Example: An AI optimizing delivery routes for a logistics company
4. **Perception:** The ability to scan the environment and interpret sensory data.
    - Example: Facial recognition systems identifying individuals
    - Example: Self-driving cars detecting pedestrians and obstacles
5. **Language Understanding:** The ability to comprehend, interpret, and generate human language.
    - Example: ChatGPT understanding and responding to your questions
    - Example: Google Translate converting text between languages

2.1.3 The "Digital Brain" Analogy

To understand AI intuitively, think of it as a "digital brain":

Kenyan Context

Kenyan Example: Just as a *mama mboga* learns to recognize regular customers, remember their preferences, and predict what they might buy today, an AI system learns from data, identifies patterns, and makes predictions.

Human Brain	AI System
Takes in information through five senses	Takes in **data** from various sources
Processes information based on experience	Processes data using **algorithms**
Makes decisions and takes actions	Produces **output** or predictions
Learns from outcomes and adjusts	Improves through **training** and feedback

Table 2.1: Human Brain vs AI System Comparison

Everyday AI Examples

- **Smartphone Face Unlock:** Your phone has "learned" the unique features of your face and compares new camera input against that stored pattern to grant or deny access.
- **M-Pesa Fraud Detection:** The system learns your typical transaction patterns and flags unusual activity (like a large transfer at 3 AM to an unknown number).
- **TikTok "For You" Page:** The algorithm learns your viewing preferences and serves content you're likely to enjoy.

2.2 Machine Learning Demystified

2.2.1 Definition and Core Concept

Machine Learning (ML): A subset of AI that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. ML focuses on developing computer programs that can access data and use it to learn for themselves.

2.2.2 The Paradigm Shift

The fundamental difference between traditional programming and machine learning:

Traditional Programming	Machine Learning
Input: Data + Rules	**Input:** Data + Answers
Process: Computer executes rules	**Process:** Computer finds patterns
Output: Answers	**Output:** Rules (Model)
Human Role: Write all rules	**Human Role:** Provide examples

Table 2.2: Traditional Programming vs Machine Learning

2.2.3 The Three Main Types of Machine Learning

1. Supervised Learning

Example

Concept: Learning with a teacher. The algorithm learns from labeled training data, where each example is paired with the correct answer.

How it works:

1. You provide the algorithm with training data
2. Each data point includes both the input and the correct output (label)
3. The algorithm learns the mapping from input to output
4. Once trained, it can predict outputs for new, unseen inputs

Real-World Examples:

- **Email Spam Detection:**
  - Training data: Thousands of emails labeled "spam" or "not spam"
  - The algorithm learns patterns that distinguish spam
  - New emails are automatically classified
- **House Price Prediction:**
  - Training data: Historical sales with features (size, location, bedrooms) and prices
  - The algorithm learns relationships between features and price
  - Predicts prices for new houses
- **Medical Diagnosis:**
  - Training data: Patient symptoms and confirmed diagnoses
  - The algorithm learns symptom-disease relationships

  - Assists doctors in diagnosing new patients

Common Algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forests
- Support Vector Machines (SVM)
- Neural Networks

2. Unsupervised Learning

Example

Concept: Learning without a teacher. The algorithm explores unlabeled data to find hidden patterns or intrinsic structures.

How it works:

1. You provide the algorithm with data (no labels)
2. The algorithm explores the data structure
3. It identifies patterns, groupings, or relationships
4. No "correct" answer exists—the algorithm discovers insights

Real-World Examples:

- **Customer Segmentation:**
  - Data: Customer purchase history, demographics, behavior
  - The algorithm groups similar customers together
  - Result: Distinct customer segments for targeted marketing
- **Market Basket Analysis:**
  - Data: Transaction records from a supermarket
  - The algorithm finds products frequently bought together
  - Result: "Customers who bought X also bought Y"
- **Anomaly Detection:**
  - Data: Network traffic patterns
  - The algorithm learns normal behavior
  - Flags unusual patterns (potential cyber attacks)

Common Algorithms:

- K-Means Clustering
- Hierarchical Clustering
- Principal Component Analysis (PCA)
- Association Rules (Apriori)
- Autoencoders

3. Reinforcement Learning

Example

Concept: Learning through trial and error. The algorithm (agent) learns to make decisions by interacting with an environment, receiving rewards or penalties.

How it works:

1. An agent interacts with an environment
2. The agent takes actions
3. The environment provides feedback (reward or penalty)
4. The agent learns to maximize cumulative rewards over time

Real-World Examples:

- **Game Playing AI:**
  - AlphaGo learned to play Go at superhuman level
  - Received rewards for winning, penalties for losing
  - Discovered strategies never seen before
- **Robotics:**
  - Robot learns to walk through trial and error
  - Reward: Moving forward without falling
  - Penalty: Falling or inefficient movement
- **Autonomous Vehicles:**
  - Car learns optimal driving strategies
  - Reward: Safe, efficient navigation
  - Penalty: Collisions, traffic violations

2.2.4 The Machine Learning Pipeline

Understanding how ML transforms data into decisions:

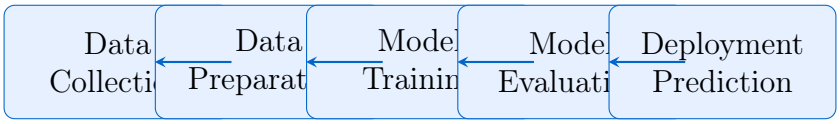


Figure 2.1: The Machine Learning Pipeline

1. **Data Collection:** Gathering relevant, high-quality data
2. **Data Preparation:** Cleaning, transforming, and organizing data
3. **Model Training:** Algorithm learns patterns from training data
4. **Model Evaluation:** Testing performance on unseen data
5. **Deployment:** Using the trained model to make predictions

2.3 Deep Learning Explained

2.3.1 Definition and Relationship to ML

Deep Learning (DL): A specialized subset of machine learning based on artificial neural networks with representation learning. The term "deep" refers to the number of layers through which data is transformed.

2.3.2 Biological Inspiration

Deep learning is inspired by the structure and function of the human brain:

Human Brain	**Artificial Neural Network**
Neurons (nerve cells)	Nodes/Artificial neurons
Synapses (connections)	Weights (connection strengths)
Neural pathways	Network architecture
Learning through experience	Training through data

Table 2.3: Biological vs Artificial Neural Networks

2.3.3 How Deep Learning Works

Layer Architecture

A deep neural network consists of multiple layers:

1. **Input Layer:** Receives the raw data
2. **Hidden Layers:** Multiple intermediate layers that process information
3. **Output Layer:** Produces the final prediction or classification

Kenyan Context

Image Recognition Example: When identifying a *mandazi* in a photo:

- Layer 1: Detects edges and basic shapes
- Layer 2: Recognizes patterns like circles and textures
- Layer 3: Identifies food-like objects
- Layer 4: Classifies as "mandazi" vs other foods

2.3.4 Why "Deep"?

The "depth" refers to the number of hidden layers:

- **Traditional Neural Networks:** 2-3 hidden layers
- **Deep Neural Networks:** Dozens to hundreds of layers
- **State-of-the-Art Models:** Can have thousands of layers

2.3.5 Applications of Deep Learning

Application	How DL is Used
Large Language Models	ChatGPT, Gemini understand and generate human language
Computer Vision	Image recognition, object detection, facial recognition
Speech Recognition	Converting spoken words to text (Siri, Google Assistant)
Autonomous Vehicles	Processing camera and sensor data for self-driving
Medical Imaging	Detecting diseases in X-rays, MRIs, CT scans
Recommendation Systems	Netflix, YouTube, Amazon recommendations

Table 2.4: Deep Learning Applications

2.4 The AI Family Tree: Visualizing Relationships

2.4.1 Understanding the Hierarchy

The relationship between AI, ML, and DL can be visualized as concentric circles:

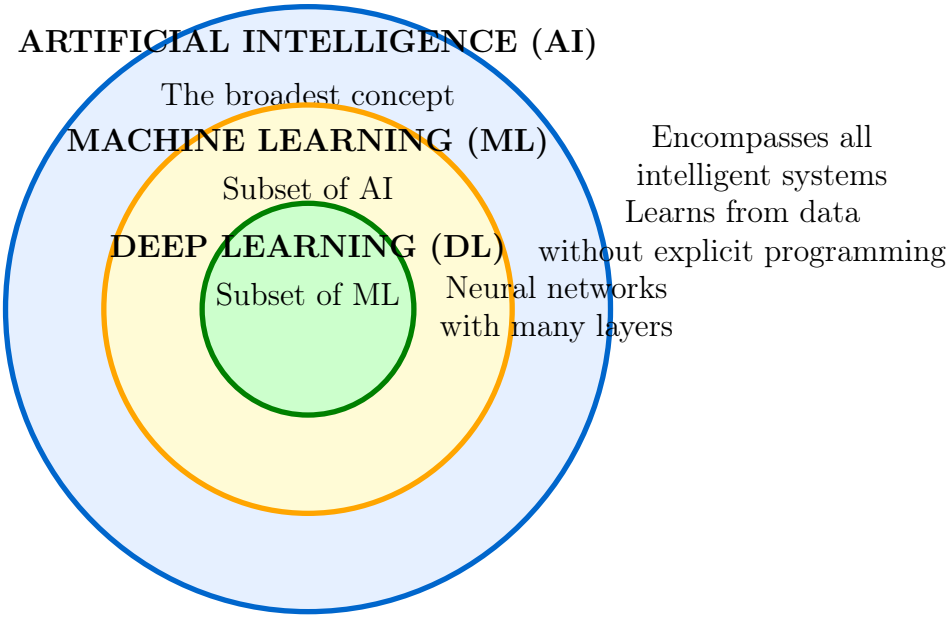


Figure 2.2: The AI Family Tree: Hierarchical Relationship

2.4.2 Key Distinctions

Feature	AI	ML	DL
Scope	Broadest	Subset of AI	Subset of ML
Approach	Rule-based or learning	Pattern recognition	Neural networks
Data Needs	Varies	Moderate	Very large
Complexity	Simple to complex	Moderate	Highly complex
Human Intervention	High	Moderate	Low
Examples	Chess programs, expert systems	Spam filters, recommendation engines	ChatGPT, image recognition

Table 2.5: Comparing AI, ML, and DL

2.5 Traditional Programming vs Machine Learning: The Fruit Sorting Analogy

2.5.1 The Scenario

Imagine you need to build a system to separate **apples** from **tomatoes** on a fast-moving conveyor belt in a packing facility.

2.5.2 Approach 1: Traditional (Rule-Based) Programming

Example

The Programmer's Task: Write explicit rules to distinguish apples from tomatoes.

Step-by-Step Process:

1. **Analyze the Problem:** What makes an apple different from a tomato?
   - Color: Both can be red
   - Shape: Both are roughly round
   - Size: Similar ranges
   - Texture: Slightly different
   - Stem: Both have stems
2. **Write Rules:**

```
IF color == "red" AND
   shape == "round" AND
   size > 5cm AND
   texture == "smooth" AND
   stem_present == True:
    CLASSIFY as "Apple"
ELSE:
    CLASSIFY as "Tomato"

```

3. **Test the System:**
   - Red apple → Correctly classified
   - Green apple → Misclassified as tomato
   - Yellow tomato → Misclassified as apple
   - Bruised apple → Misclassified
4. **Update Rules:**

```
IF color in ["red", "green", "yellow"] AND
   shape == "round" AND
   size in [5cm, 10cm] AND
   texture in ["smooth", "slightly_rough"] AND
   stem_present == True AND
   density > 0.6:
    CLASSIFY as "Apple"
ELSE:
    CLASSIFY as "Tomato"

```

The Limitations:

- **Brittle:** Fails when encountering variations not explicitly programmed
- **Maintenance Heavy:** Requires constant manual updates
- **Scalability Issues:** Becomes unmanageable as variations increase
- **Human Bias:** Limited by programmer's understanding and assumptions
- **Time-Consuming:** Each new variation requires code changes

2.5.3 Approach 2: Machine Learning (Pattern-Based)

Example

The ML Approach: Let the algorithm discover the distinguishing patterns itself.

Step-by-Step Process:

1. **Collect Training Data:**
   - Gather 10,000 images of apples (various types, colors, conditions)
   - Gather 10,000 images of tomatoes (various types, colors, conditions)
   - Label each image correctly
2. **Train the Model:**

```
# Provide data and let ML algorithm learn
training_data = load_images("apples_and_tomatoes/")
model = MachineLearningAlgorithm()
model.train(training_data)
# The algorithm discovers patterns automatically

```

3. **What the ML Algorithm Discovers:**
   - Subtle color gradients and distributions
   - Texture patterns invisible to human eyes
   - Shape variations and proportions
   - Stem characteristics
   - Shadow and lighting patterns
   - Surface reflectivity differences
4. **Test the System:**
   - Red apple → Correctly classified
   - Green apple → Correctly classified
   - Yellow tomato → Correctly classified
   - Bruised apple → Correctly classified
   - Partially hidden fruit → Correctly classified
5. **Adapt to New Data:**
   - Introduce 1,000 images of a new apple variety
   - Retrain the model
   - System automatically learns new patterns
   - No code changes required!

The Advantages:

- **Adaptive:** Learns from new data automatically
- **Robust:** Handles variations and edge cases
- **Scalable:** Improves with more data
- **Efficient:** Discovers patterns humans might miss
- **Maintainable:** Self-improving with minimal intervention

2.5.4 Side-by-Side Comparison

Aspect	Traditional Programming	Machine Learning
Development Time	Weeks to months of coding	Days to train model
Accuracy	Limited by rules	Improves with data
Handling New Cases	Requires code update	Automatic adaptation
Complexity	Grows exponentially	Scales gracefully
Human Effort	High (constant updates)	Low (initial setup)
Best For	Well-defined, stable problems	Complex, evolving problems

Table 2.6: Traditional vs ML: Fruit Sorting Comparison

2.6 Key Takeaways

Important

Essential Concepts to Remember:

1. **AI is the Umbrella:** Artificial Intelligence is the broad field encompassing any technique that enables computers to mimic human intelligence.
2. **ML is the Engine:** Machine Learning is the current driving force behind most practical AI applications, using data and patterns rather than hard-coded rules.
3. **DL is the Turbocharger:** Deep Learning is the cutting-edge technology powering the most advanced AI systems, using neural networks with many layers.
4. **Paradigm Shift:** Traditional programming requires humans to write explicit rules. Machine Learning allows computers to learn rules from data.
5. **Data is King:** The quality and quantity of data directly determine the performance of ML and DL systems.
6. **No Mathematics Required:** You don't need to be a mathematician to understand and apply these concepts. Focus on practical application and problem-solving.

2.7 Practical Exercises

2.7.1 Exercise 1: Identify AI in Your Daily Life

List five examples of AI you interact with daily:

1. _____
2. _____
3. _____
4. _____
5. _____

For each, identify whether it uses:

- Traditional programming
- Machine Learning
- Deep Learning

2.7.2 Exercise 2: Rule-Based vs ML Thinking

Scenario: You need to detect fraudulent M-Pesa transactions.

Task A (Traditional Programming): Write three rules you would program to detect fraud.

1. _____
2. _____
3. _____

Task B (Machine Learning): What data would you provide to train an ML model?

1. _____
2. _____
3. _____

2.7.3 Exercise 3: Concept Mapping

Draw a diagram showing the relationship between:

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Neural Networks
- Supervised Learning
- Unsupervised Learning

2.8 Chapter Summary

In this chapter, we have:

- Defined Artificial Intelligence and its core capabilities
- Explained Machine Learning and its three main types
- Understood Deep Learning and neural network architecture
- Visualized the hierarchical relationship between AI, ML, and DL
- Contrasted traditional programming with machine learning through the fruit sorting analogy
- Recognized that ML learns patterns from data while traditional programming follows explicit rules

2.9 Looking Ahead

In the next chapter, we will explore how these concepts are being applied specifically in the Kenyan context, examining real-world case studies from FinTech, AgriTech, healthcare, and other sectors transforming our economy.

Chapter 3

AI in the Kenyan Context: Local Applications and Case Studies

Learning Objective

Chapter Learning Objectives:

- Understand why local context matters in AI development and deployment
- Identify real-life AI applications transforming Kenyan industries
- Analyze AI use cases in FinTech, AgriTech, HealthTech, and Customer Service
- Recognize the importance of data bias and cultural relevance
- Evaluate opportunities for AI solutions in local communities

3.1 Why Local Context Matters in AI

3.1.1 The Data Bias Challenge

Important

Critical Concept: AI models are only as good as the data they are trained on. Models trained primarily on Western data often fail to understand local nuances, languages, and contexts.

Examples of Data Bias

1. **Language Bias:**
    - Global AI models excel in English but struggle with Swahili, Sheng, and local dialects
    - Example: A chatbot trained on American customer service data may not understand "Niko na shida na bill yangu" (I have a problem with my bill)
    - Impact: Excludes millions of Kenyans who prefer communicating in local languages

2. **Geographic Bias:**
    - Weather prediction models trained on European climate data fail in East African contexts
    - Example: An AI recommending planting dates based on temperate zone data would be disastrous for a maize farmer in Uasin Gishu
    - Impact: Reduces agricultural productivity and food security
3. **Economic Bias:**
    - Credit scoring models based on formal employment and bank statements exclude the informal sector
    - Example: A *mama mboga* with daily M-Pesa transactions but no payslip is deemed "uncreditworthy"
    - Impact: Perpetuates financial exclusion
4. **Cultural Bias:**
    - Image recognition systems trained on Western faces struggle with African features
    - Example: Smartphone face unlock failing more often for people with darker skin tones
    - Impact: Creates unequal access to technology

3.1.2 The Solution: Local AI Development

Kenyan Context

Building AI for Kenya, by Kenyans: The most effective AI solutions are developed with deep understanding of local needs, contexts, and constraints.

Key Principles for Local AI:

1. **Local Data Collection:** Train models on data that represents Kenyan reality
2. **Community Involvement:** Engage end-users in the design and testing process
3. **Cultural Sensitivity:** Respect local customs, languages, and social structures
4. **Affordability:** Design solutions accessible to low-income populations
5. **Infrastructure Awareness:** Account for intermittent internet and electricity

3.2 FinTech: Revolutionizing Financial Services

3.2.1 Mobile Lending and Credit Scoring

Example

The Challenge: Over 8 million Kenyans lack access to formal credit because they have no credit history, payslips, or collateral. Traditional banks require extensive documentation that excludes the informal sector.

The AI Solution

Machine Learning algorithms analyze **alternative data sources** to assess creditworthiness:

Data Source	What It Reveals	Example
M-Pesa Transaction History	Income stability, spending patterns, financial responsibility	Regular savings transfers indicate discipline
Airtime Usage	Communication patterns, social network stability	Consistent top-ups suggest reliable income
Bill Payments	Responsibility and reliability	Timely utility payments show trustworthiness
Device Information	Stability and authenticity	Long-term device use indicates stability
Social Network Analysis	Trustworthiness through associations	Connections to reliable users boost score

Table 3.1: Alternative Data for Credit Scoring

How It Works

1. **Data Collection:** User grants permission to access mobile money and phone data
2. **Feature Extraction:** ML algorithm identifies hundreds of predictive features
    - Average daily balance
    - Frequency of transactions
    - Diversity of transaction types
    - Time between income and spending
    - Network of regular contacts
3. **Model Training:** Historical data from millions of users trains the algorithm
    - Input: User features
    - Output: Loan repayment (yes/no)
    - Algorithm learns patterns distinguishing good from bad borrowers

4. **Real-Time Scoring:** New applicants receive instant credit decisions
    - Application submitted via USSD or app
    - Algorithm analyzes data in seconds
    - Loan amount and terms determined automatically
    - Funds disbursed instantly to M-Pesa

Real-World Examples

- **M-Shwari (Safaricom & CBA):**
    - Launched in 2012, first mobile-based lending in Kenya
    - Over 25 million users to date
    - Uses ML to determine loan limits (Ksh 100 to Ksh 50,000)
    - Interest rates: 7.5% flat fee for 30 days
- **KCB M-Pesa:**
    - Partnership between KCB Bank and Safaricom
    - Loans from Ksh 100 to Ksh 1 million
    - Uses advanced ML models for risk assessment
    - Integrates with traditional banking data
- **Branch and Tala:**
    - International fintechs with strong Kenyan presence
    - Analyze smartphone data (with permission)
    - Consider app usage, contacts, SMS patterns
    - Provide loans without M-Pesa requirement

Kenyan Context

Impact: A *boda boda* rider in Nairobi can access Ksh 5,000 emergency loan within minutes to repair his motorcycle, enabling him to continue earning. Without AI-powered lending, he would need weeks to navigate bank bureaucracy.

3.2.2 Fraud Detection Algorithms

Example

The Challenge: Kenya processes over Ksh 50 trillion annually through mobile money. With this volume comes sophisticated fraud: SIM swap attacks, phishing, unauthorized transactions, and social engineering.

How AI Detects Fraud

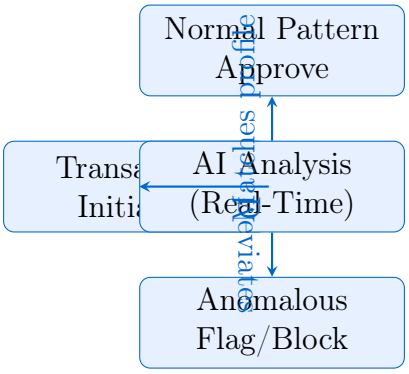


Figure 3.1: Real-Time Fraud Detection Flow

What the AI Monitors:

1. **Behavioral Patterns:**
   - Typical transaction amounts (Ksh 500-5,000)
   - Usual transaction times (9 AM - 8 PM)
   - Common recipients (family, regular businesses)
   - Frequent locations (home, work area)
2. **Anomaly Detection:**
   - **Red Flag 1:** Ksh 50,000 transfer at 3 AM (unusual time and amount)
   - **Red Flag 2:** First-time recipient with no history
   - **Red Flag 3:** Transaction from new device/location
   - **Red Flag 4:** Multiple rapid transactions (possible account takeover)
3. **Network Analysis:**
   - Identifies connections to known fraudsters
   - Detects money muling patterns
   - Flags suspicious recipient networks

Response Mechanisms

When fraud is detected:

Risk Level	AI Response
Low Risk	Transaction approved instantly
Medium Risk	Additional verification required (PIN, biometric, OTP)
High Risk	Transaction blocked, SMS alert sent, account temporarily frozen
Critical Risk	Immediate account lockdown, fraud team notified, law enforcement alerted

Table 3.2: Fraud Detection Response Levels

Kenyan Context

Real Incident: In 2024, Safaricom's AI detected a coordinated SIM swap attack targeting 500+ accounts. The system identified unusual patterns: multiple SIM change requests from the same location, followed immediately by large transfer attempts. The AI automatically blocked all suspicious transactions and alerted the fraud team, preventing losses exceeding Ksh 50 million.

3.3 Customer Service: Localized Chatbots

3.3.1 The Rise of AI-Powered Customer Support

Example

The Challenge: Kenyan businesses receive thousands of customer inquiries daily: "How do I reset my PIN?", "When will my loan be approved?", "Why was my transaction declined?" Human agents cannot handle this volume 24/7.

Business Benefits

Benefit	Impact
24/7 Availability	Customers get help anytime, even at midnight or on public holidays
Instant Response	Wait time reduced from 30 minutes to 3 seconds
Cost Reduction	One chatbot handles 10,000+ conversations simultaneously
Consistency	Every customer receives accurate, standardized information
Scalability	Handle peak periods (month-end, holidays) without hiring more staff
Multilingual Support	Serve customers in English, Swahili, and Sheng

Table 3.3: Benefits of AI Chatbots

3.3.2 Local Language Processing

Kenyan Context

Breaking Language Barriers: Advanced chatbots now understand and respond in Swahili and Sheng, making technology accessible to all Kenyans.

Examples of Local Language Understanding

Customer Query	Language	Chatbot Response
"Niko na shida na bill yangu, imeongezeka sana"	Swahili	"Nitasaidia kuchunguza bill yako. Unaweza kuniambia nambari yako ya akaunti?"
"Nahitaji loan, process ni gani?"	Sheng	"Poa! Nitakusaidia. Check kwenye app yangu, click 'Loans' then follow instructions. Una limit gani?"
"How do I reverse a wrong transaction?"	English	"I'll help you with that. Please provide the transaction code and recipient number."
"Nime-send pesa kwa nambari mbaya, nisaidie"	Swahili	"Tulia, nitakusaidia. Toa code ya transaction na nitacheck kama inaweza kurejeshwa."

Table 3.4: Multilingual Chatbot Examples

Real-World Implementations

- **Safaricom's Zuri:**
  - AI assistant on WhatsApp and Facebook Messenger
  - Handles 1 million+ queries monthly
  - Understands Swahili, English, and basic Sheng
  - Resolves 70% of queries without human intervention
  - Services: Airtime, data, M-Pesa, device support
- **Equity Bank's Eazzy Banking Chatbot:**
  - Available on mobile app and website
  - Assists with account inquiries, transfers, loan applications
  - Escalates complex issues to human agents seamlessly
  - Reduces call center volume by 40%
- **KCB's Virtual Assistant:**

- Integrated with KCB Mobile app
- Provides balance inquiries, mini statements, branch locator
- Learns from customer interactions to improve responses
- Available 24/7 in English and Swahili

3.4 AgriTech: Transforming Agriculture

3.4.1 Crop Disease Detection

Example

The Challenge: Smallholder farmers lose 30-40% of their crops to diseases and pests. Extension officers are too few to visit every farm. By the time a disease is identified, it has often spread widely.

The AI Solution: Computer Vision

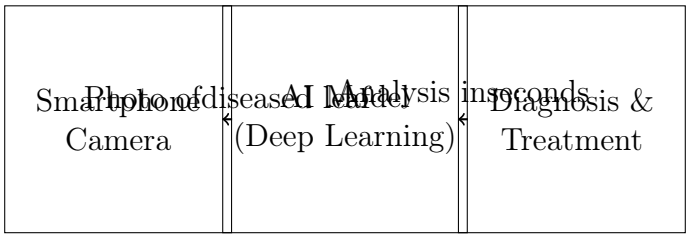


Figure 3.2: AI-Powered Crop Disease Detection Process

How It Works:

1. **Data Collection:**
   - Thousands of images of healthy and diseased crops
   - Each image labeled by agricultural experts
   - Covers multiple diseases: blight, rust, mosaic virus, etc.
   - Includes variations: different growth stages, lighting, angles
2. **Model Training:**
   - Deep Learning CNN (Convolutional Neural Network) trained on dataset
   - Learns visual patterns unique to each disease
   - Achieves 90-95% accuracy in controlled tests
3. **Farmer Usage:**
   (a) Farmer notices unusual spots on maize leaves
   (b) Opens app (e.g., PlantVillage Nuru)

(c) Takes photo of affected leaf

(d) AI analyzes image in 2-3 seconds

(e) Returns diagnosis: "Maize Lethal Necrosis (MLN) - 94% confidence"

(f) Provides treatment: "Remove infected plants, apply pesticide X, contact extension officer"

(g) Connects farmer to nearest agrovet for treatment purchase

Real-World Example: PlantVillage Nuru

Kenyan Context

PlantVillage Nuru (Nuru means "light" in Swahili) is an AI assistant helping African farmers:

- **Developed by:** Penn State University and International Institute of Tropical Agriculture
- **Crops Covered:** Cassava, maize, potato, tomato, beans
- **Languages:** English, Swahili, French, Portuguese
- **Offline Capability:** Works without internet after initial download
- **Impact:** Helped 500,000+ farmers across Africa

3.4.2 Weather Prediction and Yield Optimization

Example

The Challenge: Climate change has made rainfall patterns unpredictable. Traditional farming calendars no longer work. Farmers plant based on guesswork, often losing crops to drought or floods.

AI-Powered Precision Agriculture

AI Application	How It Helps Farmers
Hyper-Local Weather Forecasting	ML models analyze satellite data, historical patterns, and real-time sensors to predict rainfall at village level with 85% accuracy up to 14 days in advance
Optimal Planting Dates	AI recommends exact planting windows based on soil moisture, temperature, and predicted rainfall
Irrigation Scheduling	Smart systems tell farmers when and how much to water, reducing water waste by 30%
Fertilizer Optimization	AI analyzes soil composition and crop needs to recommend precise fertilizer types and quantities
Yield Prediction	Predicts harvest size 2-3 months in advance, helping farmers plan storage and sales
Pest Outbreak Alerts	Monitors weather conditions favorable for pests and sends early warnings

Table 3.5: AI Applications in Precision Agriculture

Case Study: DigiFarm by Safaricom

- **Platform:** Integrated mobile platform for smallholder farmers
- **AI Features:**
  - Personalized farming tips via SMS based on location and crop
  - Weather alerts specific to farmer's GPS coordinates
  - Market price predictions to optimize selling timing
  - Credit scoring using alternative data (farming history, M-Pesa transactions)
- **Impact:**
  - 1.5 million farmers registered
  - 25% average yield increase
  - 40% reduction in input costs through bulk purchasing
  - Access to Ksh 2 billion in agricultural loans

3.5 Other Emerging Sectors

3.5.1 HealthTech: AI in Healthcare

Kenyan Context

The Healthcare Challenge: Kenya has only 0.2 doctors per 1,000 people. Rural clinics lack specialists. Patients wait weeks for diagnosis and test results.

AI Applications in Healthcare

1. **Medical Imaging Analysis:**
   - **Problem:** Radiologists are concentrated in Nairobi; rural clinics have no specialists
   - **Solution:** AI analyzes X-rays, ultrasounds, and CT scans
   - **Example:** TB detection AI scans chest X-rays in 30 seconds with 94% accuracy
   - **Impact:** Enables early diagnosis in remote areas
2. **Disease Outbreak Prediction:**
   - ML models analyze climate data, population movement, and historical outbreaks
   - Predicts cholera, malaria, and dengue hotspots 2-4 weeks in advance
   - Enables proactive resource allocation (vaccines, medications, personnel)
3. **Telemedicine Triage:**
   - AI chatbot asks symptom questions
   - Determines urgency: emergency, clinic visit, or home care
   - Routes patients to appropriate care level
   - Reduces unnecessary hospital visits

3.5.2 EdTech: Personalized Learning

Example

The Challenge: Kenyan classrooms have 50-70 students per teacher. Individual attention is impossible. Students fall behind without timely intervention.

AI-Powered Education

- **Adaptive Learning Platforms:**
  - AI assesses student's current knowledge level
  - Creates personalized learning path

  - Adjusts difficulty based on performance
  - Identifies knowledge gaps and provides targeted practice

- **Automated Grading:**
  - AI grades multiple-choice, fill-in-blank, and even essay questions
  - Provides instant feedback to students
  - Frees teachers to focus on instruction rather than marking

- **Early Warning Systems:**
  - Analyzes attendance, assignment completion, test scores
  - Flags students at risk of failing
  - Alerts teachers and parents for early intervention

3.5.3 Logistics and Transportation

Kenyan Context

The Nairobi Traffic Challenge: Nairobi loses Ksh 50 billion annually to traffic congestion. Delivery companies struggle with unpredictable travel times.

AI Solutions

1. **Route Optimization:**
   - ML analyzes historical traffic patterns, weather, events, time of day
   - Predicts optimal routes in real-time
   - Reduces delivery times by 20-30%
   - Saves fuel and reduces emissions
2. **Demand Forecasting:**
   - Predicts delivery demand by area and time
   - Optimizes driver and vehicle allocation
   - Reduces idle time and improves efficiency
3. **Real-World Example: Sendy**
   - Logistics platform connecting businesses with drivers
   - AI matches deliveries with nearest available driver
   - Dynamic pricing based on demand, distance, traffic
   - Reduced delivery costs by 30% for SMEs

3.6 Group Discussion Framework

3.6.1 Identifying AI Opportunities in Your Community

Use this framework to brainstorm AI solutions:

AI Solution Design Canvas

1. **Problem Identification:**
   - What specific challenge exists in your community/business?
   - Who is affected and how?
   - How often does it occur?
   - What is the economic/social impact?

2. **Current Solutions:**
   - How is this problem currently addressed?
   - What are the limitations?
   - Why are current solutions inadequate?

3. **AI Applicability:**
   - Could AI automate or improve this process?
   - What type of AI would be suitable?
     - Computer Vision (images/video)
     - Natural Language Processing (text/voice)
     - Predictive Analytics (forecasting)
     - Recommendation Systems (personalization)

4. **Data Requirements:**
   - What data would the AI need?
   - Is this data available?
   - How would you collect it?
   - Are there privacy concerns?

5. **Feasibility Assessment:**
   - Technical feasibility: Can it be built with current tools?
   - Economic feasibility: Is it affordable?
   - Cultural acceptance: Will people use it?
   - Infrastructure: Does supporting infrastructure exist?

3.6.2 Case Study Analysis Exercise

Example

Group Activity: Analyze the following scenario and propose an AI solution.

Scenario: A rural county hospital in Kenya faces these challenges:

- Only 2 doctors for 50,000 residents
- Patients travel 50+ km for basic consultations
- Long wait times (3-5 hours average)
- High rate of preventable diseases
- Limited medical supplies due to poor inventory management
- No specialist access (cardiology, dermatology, etc.)

Your Task:

1. Identify 2-3 AI solutions that could address these challenges
2. For each solution, specify:
    - Type of AI technology
    - Data requirements
    - Implementation steps
    - Expected impact
    - Potential challenges
3. Present your solution to the group (5 minutes)

3.7 Chapter Summary

- **Local Context is Critical:** AI models trained on Western data often fail in Kenyan contexts due to language, cultural, and infrastructure differences.
- **FinTech Leadership:** Kenya leads Africa in AI-powered mobile lending and fraud detection, using alternative data to serve the unbanked.
- **Customer Service Transformation:** Localized chatbots understanding Swahili and Sheng are making digital services accessible to all Kenyans.
- **AgriTech Innovation:** Computer vision for disease detection and ML for weather prediction are helping smallholder farmers increase yields and reduce losses.
- **Emerging Opportunities:** HealthTech, EdTech, and logistics are ripe for AI disruption, with significant social and economic impact potential.
- **Ethical Imperative:** AI development must prioritize fairness, transparency, and inclusion to avoid perpetuating existing inequalities.

3.8 Practical Exercises

3.8.1 Exercise 1: AI Audit

Identify 5 AI applications you interact with weekly. For each:

- Name the application
- Describe what it does
- Identify the type of AI (ML, DL, NLP, Computer Vision)
- Hypothesize what data it uses
- Assess whether it's designed with Kenyan context in mind

3.8.2 Exercise 2: Local Problem Analysis

Choose a problem in your community:

- Describe the problem in detail
- Identify who is affected and how
- Research if AI solutions exist elsewhere
- Propose how it could be adapted for local context
- Identify potential data sources
- List implementation challenges

3.8.3 Exercise 3: Bias Detection

Analyze a popular AI tool (e.g., ChatGPT, Google Translate):

- Test it with Kenyan-specific queries
- Identify where it fails or provides inaccurate information
- Document examples of cultural or linguistic bias
- Propose improvements for local relevance

Chapter 4

Navigating the AI Tool Landscape

Learning Objective

Chapter Learning Objectives:

- Understand the major categories of AI tools available
- Compare popular Generative AI platforms (ChatGPT, Gemini, Copilot, Qwen, DeepSeek)
- Evaluate No-Code ML platforms (Teachable Machine, AutoML, Azure ML)
- Select appropriate tools for specific tasks and contexts
- Understand tool limitations, costs, and ethical considerations

4.1 Introduction to AI Tools

4.1.1 The AI Tool Ecosystem

The AI landscape has exploded with hundreds of tools. Understanding categories helps you choose the right tool:

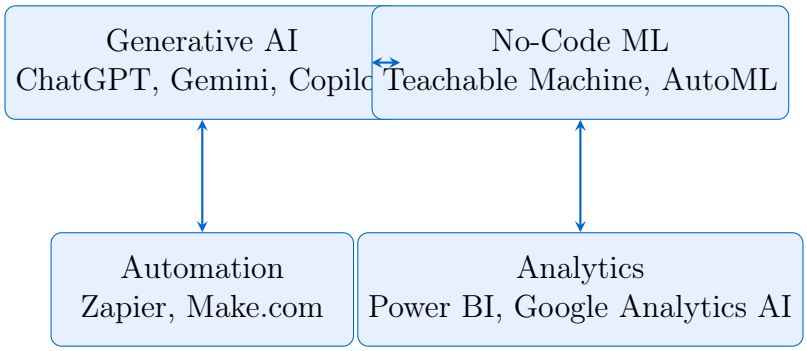


Figure 4.1: AI Tool Categories and Relationships

4.1.2 Tool Selection Criteria

When choosing an AI tool, consider:

Criterion	Questions to Ask
Task Fit	Does this tool solve my specific problem? What are its strengths and weaknesses?
Ease of Use	Do I need coding skills? What is the learning curve?
Cost	Is there a free tier? What are the pricing tiers? Is it sustainable for my budget?
Data Privacy	Where is my data stored? Who owns it? Is it used for training?
Local Relevance	Does it support Swahili/Sheng? Does it understand Kenyan context?
Integration	Does it work with tools I already use (Google Workspace, Microsoft 365)?
Reliability	What is the uptime? Is technical support available?

Table 4.1: AI Tool Selection Framework

4.2 Generative AI Platforms

4.2.1 ChatGPT (OpenAI)

Example

Overview: ChatGPT is the most popular conversational AI, launched in November 2022. It excels at natural language understanding and generation.

Key Features

- **Conversational Interface:** Chat naturally in English, Swahili, or other languages
- **Text Generation:** Write emails, reports, essays, code, poetry
- **Question Answering:** Get explanations on complex topics
- **Summarization:** Condense long documents into key points
- **Translation:** Translate between 50+ languages
- **Code Assistance:** Write, debug, and explain code in multiple languages

- **Creative Writing:** Generate stories, marketing copy, scripts

Versions and Pricing

Version	Features	Cost
GPT-3.5 (Free)	Basic chat, text generation, limited knowledge (2023 cutoff)	Free
GPT-4 (Plus)	Advanced reasoning, larger context, faster responses, web browsing	$20/month
GPT-4o (Omni)	Multimodal (text + images + voice), real-time conversation	Included in Plus
Team/Business	Shared workspace, admin controls, data privacy guarantees	$25-30/user/month

Table 4.2: ChatGPT Versions and Pricing

Best Use Cases

- Drafting professional emails and documents
- Brainstorming ideas and solutions
- Learning new concepts with explanations
- Writing and debugging code
- Creating content (blog posts, social media, marketing)
- Research assistance and summarization

Limitations

- Knowledge cutoff (may not know recent events)
- Can generate plausible but incorrect information ("hallucinations")
- Limited Swahili/Sheng understanding compared to English
- Free version has usage limits during peak times
- No native integration with Kenyan-specific data sources

4.2.2 Gemini (Google)

Example

Overview: Google's AI assistant, formerly Bard, deeply integrated with Google's ecosystem and services.

Key Features

- **Google Integration:** Access Gmail, Drive, Docs, Sheets directly
- **Real-Time Information:** Live web search with citations
- **Multimodal:** Analyze images, PDFs, documents
- **Workspace Integration:** Draft emails in Gmail, create docs in Google Docs
- **Code Execution:** Write and run Python code in sandbox
- **Image Generation:** Create images from text descriptions

Advantages Over ChatGPT

Feature	Advantage
Real-Time Data	Always connected to live web; knows current events
Google Ecosystem	Seamless integration with tools you already use
Free Access	Advanced features available at no cost
Citations	Provides sources for information
Document Analysis	Can read and summarize uploaded PDFs, Word docs

Table 4.3: Gemini Advantages

Best Use Cases

- Research requiring current information
- Working within Google Workspace
- Analyzing documents and extracting insights
- Creating content with image support
- Academic writing with citations

4.2.3 Microsoft Copilot

Example

Overview: Microsoft's AI assistant built on GPT-4, integrated with Windows and Microsoft 365.

Key Features

- **Windows Integration:** Built into Windows 11 taskbar
- **Microsoft 365 Copilot:** Works in Word, Excel, PowerPoint, Outlook, Teams
- **Enterprise Features:** Data privacy, compliance, security
- **Code Generation:** GitHub Copilot for developers
- **Business Intelligence:** Analyze data in Excel with natural language

Microsoft 365 Copilot Capabilities

Application	AI Capabilities
Word	Draft documents, summarize long texts, rewrite sections, check tone
Excel	Analyze data, create formulas, generate charts, identify trends
PowerPoint	Create presentations from outline, design slides, add speaker notes
Outlook	Draft emails, summarize threads, suggest replies, schedule meetings
Teams	Summarize meetings, extract action items, answer questions about discussion

Table 4.4: Microsoft 365 Copilot Features

Pricing

- **Copilot (Free):** Web-based chat, basic features
- **Copilot Pro:** $20/month - Priority access, faster responses, image creation
- **Microsoft 365 Copilot:** $30/user/month - Full Office integration (requires Microsoft 365 subscription)

4.2.4 Qwen (Alibaba)

Kenyan Context

Overview: Qwen is a powerful open-source AI model from Alibaba Cloud, with strong multilingual capabilities including excellent support for African languages.

Key Features

- **Multilingual Excellence:** Strong performance in Swahili, Arabic, Chinese
- **Open Source:** Models available for local deployment
- **Long Context:** Can process very long documents (up to 128K tokens)
- **Code Generation:** Competitive with GPT-4 in programming tasks
- **Reasoning:** Strong mathematical and logical reasoning
- **Vision-Language:** Qwen-VL understands images and text together

Advantages for Kenyan Users

- Better Swahili understanding than Western models
- Free and open-source options available
- Can be deployed locally for data privacy
- Strong performance on African language tasks
- No usage limits or subscription fees (self-hosted)

Access Methods

1. **Web Interface:** chat.qwen.ai (free)
2. **API:** For developers integrating into applications
3. **Local Deployment:** Download models for offline use (requires technical skills)
4. **Hugging Face:** Community-hosted versions

4.2.5 DeepSeek

Example

Overview: DeepSeek is a Chinese AI company known for highly efficient models with strong reasoning and coding capabilities, offered at very competitive prices.

Key Features

- **Advanced Reasoning:** Excels at complex problem-solving
- **Coding Expertise:** Strong performance on programming tasks
- **Cost-Effective:** API pricing 90% cheaper than GPT-4
- **Long Context:** 128K token context window
- **Open Weights:** Models available for research and commercial use
- **Multilingual:** Supports 100+ languages

Best For

- Developers needing coding assistance
- Businesses requiring cost-effective AI at scale
- Researchers wanting open models
- Applications requiring complex reasoning

4.2.6 Manus

Example

Overview: Manus is an AI agent platform designed to perform multi-step, autonomous tasks across different applications and workflows.

Key Features

- **Autonomous Agents:** AI that can complete tasks without constant supervision
- **Multi-Step Workflows:** Handles complex sequences of actions
- **Tool Integration:** Connects to web browsers, APIs, databases
- **Research Automation:** Can conduct web research, compile reports
- **Data Analysis:** Extract, clean, and analyze data from multiple sources

Use Case Example

Task: "Research the top 10 agritech startups in Kenya, compile their funding information, and create a comparison spreadsheet."

Manus Workflow:

1. Searches web for Kenyan agritech startups
2. Visits each company website
3. Extracts funding data from Crunchbase, news articles

4. Compiles information into structured format
5. Creates Google Sheets spreadsheet with comparison
6. Emails you the link

4.3 No-Code Machine Learning Platforms

4.3.1 Google Teachable Machine

Kenyan Context

Overview: Teachable Machine is the easiest way to learn how machine learning works. No coding required. Train models in your browser using your webcam, microphone, or uploaded files.

How It Works

1. **Choose Project Type:**
   - Image Project (classify images)
   - Audio Project (classify sounds)
   - Pose Project (classify body positions)
2. **Collect Data:**
   - Use webcam to capture images (e.g., 50 photos of "happy" face, 50 of "sad" face)
   - Or upload existing images from your computer
   - Organize into classes (categories)
3. **Train Model:**
   - Click "Train Model" button
   - Wait 30-60 seconds
   - Model learns patterns automatically
4. **Test Model:**
   - Use webcam or upload test images
   - See real-time predictions
   - Check accuracy
5. **Export Model:**
   - Download for use in websites, apps
   - Share via link
   - Integrate with other tools

Practical Examples

- **Mango Ripeness Classifier:**
  - Class 1: Green (unripe) mangoes
  - Class 2: Yellow (ripe) mangoes
  - Class 3: Brown (overripe) mangoes
  - Use: Help farmers sort harvest
- **Hand Gesture Controller:**
  - Train model to recognize hand signals
  - Control presentation slides with gestures
  - Accessibility tool for disabled users
- **Sound Classifier:**
  - Detect machine sounds
  - Alert maintenance before breakdown
  - Predictive maintenance for factories

Advantages

- Completely free
- No coding required
- Runs in browser (no installation)
- Fast training (under 1 minute)
- Great for learning ML concepts
- Export models for real applications

Limitations

- Simple models only (not for complex tasks)
- Limited to 3 project types (image, audio, pose)
- No text classification
- Models not as accurate as professional tools
- Best for prototyping and education

4.3.2 Google Cloud AutoML

Example

Overview: AutoML enables developers with limited machine learning expertise to train high-quality models tailored to their specific business needs.

AutoML Product Suite

Product	Use Case
AutoML Vision	Custom image classification (e.g., detect crop diseases, product defects)
AutoML Natural Language	Text classification, sentiment analysis (e.g., categorize customer feedback)
AutoML Tables	Structured data prediction (e.g., sales forecasting, churn prediction)
AutoML Video Intelligence	Analyze video content (e.g., security footage analysis)
AutoML Translation	Custom translation models for specific domains

Table 4.5: Google Cloud AutoML Products

Workflow

1. **Prepare Data:**
   - Collect labeled dataset (e.g., 1,000 images with correct labels)
   - Upload to Google Cloud Storage
   - Ensure data quality and balance
2. **Create Dataset:**
   - In AutoML interface, create new dataset data from Cloud Storage
   - Verify labels and organization
3. **Train Model:**
   - Click "Train" button
   - Choose training budget (higher = better accuracy, more cost)
   - Training takes 1-4 hours typically
   - Google handles all complexity automatically
4. **Evaluate Model:**
   - Review accuracy metrics
   - Check precision, recall, F1 score
   - Identify weak classes
   - Iterate if needed (add more data, retrain)
5. **Deploy Model:**

- Deploy to cloud endpoint
- Get API key
- Integrate into your application
- Pay per prediction (typically $1-5 per 1,000 predictions)

Pricing

- **Training:** $3-10 per hour (depending on model type)
- **Deployment:** $1.50-3 per hour (while model is online)
- **Predictions:** $1-5 per 1,000 predictions
- **Free Tier:** $300 credit for new Google Cloud users

Best For

- Businesses needing custom AI models
- Projects requiring higher accuracy than Teachable Machine
- Production applications with API integration
- Teams without ML expertise but with technical developers

4.3.3 Microsoft Azure Machine Learning

Example

Overview: Azure ML is Microsoft's enterprise-grade cloud platform for building, training, and deploying machine learning models at scale.

Key Features

- **Drag-and-Drop Designer:** Build models visually without coding
- **Automated ML:** Automatically find best algorithm and parameters
- **Jupyter Notebooks:** Code-based development for data scientists
- **MLOps:** Version control, monitoring, retraining automation
- **Enterprise Security:** Compliance, data encryption, access controls
- **Integration:** Works with Power BI, Excel, Azure services

Automated ML Workflow

1. Upload dataset (CSV, Excel, database)
2. Select target column (what you want to predict)
3. Choose task type (classification, regression, forecasting)
4. Configure compute resources
5. Click "Run" - Azure tests dozens of algorithms automatically
6. Review results and select best model
7. Deploy as web service API

Pricing

- **Compute:** Pay for compute time used (starting at $0.05/hour)
- **Storage:** Standard Azure storage rates
- **Deployment:** Based on instance size ($50-500/month)
- **Free Tier:** Limited free compute for learning

4.4 AI Automation Tools

4.4.1 Zapier

Example

Overview: Zapier connects your apps and automates workflows without coding. Trigger actions in one app based on events in another.

How It Works

Concept: "When THIS happens, do THAT"

- **Trigger:** Event that starts the automation
- **Action:** What happens when trigger occurs
- **Zap:** One complete automation workflow

Example: AI-Powered Lead Management

Goal: Automatically qualify and respond to new customer inquiries.

Zap Workflow:

1. **Trigger:** New form submission on website
2. **Action 1:** Send form data to ChatGPT via API
   - Prompt: "Analyze this inquiry and rate lead quality 1-10"
   - Get AI response with score and summary
3. **Action 2:** Add lead to Google Sheets with AI score
4. **Action 3:** If score > 7, send personalized email via Gmail
5. **Action 4:** If score < 5, add to "nurture" email list
6. **Action 5:** Notify sales team on Slack for high-value leads

Pricing

- **Free:** 100 tasks/month, 5 Zaps, 15-minute update time
- **Starter:** $20/month - 750 tasks, 20 Zaps, 5-minute updates
- **Professional:** $50/month - 2,000 tasks, unlimited Zaps, 2-minute updates
- **Team:** $100/month - Collaboration features

4.4.2 Make.com (formerly Integromat)

Example

Overview: Similar to Zapier but with more advanced visual workflow builder and lower pricing for high-volume automation.

Advantages Over Zapier

- More affordable for high-volume usage
- Visual workflow builder (drag-and-drop)
- More complex logic and data manipulation
- Better for technical users
- Stronger API integration capabilities

Use Case: Automated Content Creation

1. Monitor RSS feed for industry news
2. When new article published:
    - Extract article text
    - Send to AI for summarization
    - Generate social media post from summary
    - Create image using AI image generator
    - Schedule post on Twitter, LinkedIn, Facebook
    - Save to content calendar in Google Sheets

4.5 Tool Comparison Matrix

Tool	Free Tier	Ease of Use	Swahili Support	Best For	Coding Required
ChatGPT	Yes			General tasks, writing	No
Gemini	Yes			Research, Google integration	No
Copilot	Yes			Microsoft 365 users	No
Qwen	Yes			Multilingual, open-source	Optional
DeepSeek	Yes			Coding, reasoning	Optional
Teachable Machine	Yes		N/A	Learning ML basics	No
AutoML	$300 credit		N/A	Production ML models	Some
Azure ML	Limited		N/A	Enterprise ML	Some
Zapier	Yes		N/A	Simple automations	No
Make.com	Yes		N/A	Complex workflows	No

Table 4.6: AI Tool Comparison Matrix

4.6 Choosing the Right Tool

4.6.1 Decision Framework

Tool Selection Decision Tree

Question 1: What is your primary goal?

① Generate text/content → Go to Question 2

② Build ML model → Go to Question 3

③ Automate workflows → Go to Question 4

Question 2: Do you need real-time web information?

- Yes → **Gemini** (free, live data)
- No → **ChatGPT** (best overall quality)
- Budget constraint → **Qwen** (free, open-source)

Question 3: What is your technical skill level?

- Beginner → **Teachable Machine** (no code, instant results)
- Intermediate → **AutoML** (some setup, production-ready)
- Advanced → **Azure ML** or custom code (maximum control)

Question 4: How complex is your workflow?

- Simple (1-2 steps) → **Zapier** (easiest)
- Complex (multiple branches) → **Make.com** (more powerful)
- Enterprise scale → Custom API integration

4.7 Chapter Summary

- **Generative AI:** ChatGPT, Gemini, Copilot, Qwen, and DeepSeek each have unique strengths. Choose based on your specific needs, budget, and language requirements.
- **No-Code ML:** Teachable Machine is perfect for learning and prototyping. AutoML and Azure ML are for production applications requiring higher accuracy and reliability.
- **Automation Tools:** Zapier and Make.com enable powerful workflows connecting AI with your existing apps, saving hours of manual work.
- **Tool Selection:** There is no "best" tool—only the best tool for your specific use case. Consider cost, ease of use, language support, and integration needs.

- **Start Simple:** Begin with free tools to learn concepts before investing in paid platforms.
- **Combine Tools:** The most powerful solutions often combine multiple tools (e.g., ChatGPT + Zapier + Google Sheets).

4.8 Practical Exercises

4.8.1 Exercise 1: Tool Exploration

Spend 30 minutes with each of these tools:

1. ChatGPT (chat.openai.com)
2. Gemini (gemini.google.com)
3. Teachable Machine (teachablemachine.withgoogle.com)

For each tool, document:

- Three tasks you performed
- What worked well
- What was challenging
- One potential use case for your work/community

4.8.2 Exercise 2: Build a Simple Automation

Using Zapier free tier or Make.com:

1. Create an automation that:
    - Monitors your email for specific keyword
    - When found, sends summary to ChatGPT
    - Saves AI analysis to Google Sheets
2. Test with 3 different emails
3. Document the workflow with screenshots

4.8.3 Exercise 3: Train a Custom Model

Using Teachable Machine:

1. Create an image classification project
2. Choose a locally relevant problem (e.g., classify fruits, detect emotions, identify objects)
3. Collect at least 30 images per class

4. Train the model
5. Test accuracy with new images
6. Export and share your model

Chapter 5

The Art of Prompt Engineering

Learning Objective

Chapter Learning Objectives:

- Understand what prompt engineering is and why it matters
- Master the four components of effective prompts
- Apply iterative prompting techniques
- Use advanced prompting strategies (few-shot, chain-of-thought)
- Avoid common prompting mistakes
- Create prompt templates for repeatable tasks

5.1 Introduction to Prompt Engineering

5.1.1 What is Prompt Engineering?

Definition: Prompt engineering is the practice of designing and refining input instructions (prompts) to guide AI models toward producing desired, high-quality outputs.

Why It Matters

Example

The Difference Good Prompting Makes:

Poor Prompt:

```
"Write about climate change"
```

Result: Generic, unfocused 500-word essay covering basic facts you already know.

Excellent Prompt:

```
"Act as a climate scientist speaking to Kenyan farmers. Explain how climate
change specifically affects maize production in Rift Valley, focusing on:
1. Changes in rainfall patterns (2020-2026)
2. Impact on planting seasons
3. Three practical adaptation strategies
Keep it under 400 words, use simple language, and include local examples."
```

Result: Targeted, actionable information directly relevant to your audience.

5.1.2 The GIGO Principle

Important

Garbage In, Garbage Out (GIGO): The quality of AI output is directly proportional to the quality of your input. Vague, unclear, or incomplete prompts produce poor results.

5.2 The Four Components of Effective Prompts

5.2.1 1. Role (Persona)

Purpose: Tell the AI who it should pretend to be. This sets the tone, expertise level, and perspective.

Role Examples	Effect on Output
"Act as an experienced marketing manager"	Professional, strategic, business-focused
"Act as a patient teacher explaining to a beginner"	Simple, step-by-step, encouraging
"Act as a skeptical investor"	Critical, focused on ROI and risks
"Act as a creative storyteller"	Engaging, vivid, emotional
"Act as a data scientist"	Technical, evidence-based, precise

Table 5.1: Role Component Examples

Examples:

- **Weak:** "Write an email to a client"
- **Strong:** "Act as a senior customer success manager with 10 years of experience. Write a professional email to a frustrated client whose order was delayed."

5.2.2 2. Context (Background Information)

Purpose: Provide the AI with necessary background to understand the situation and generate relevant responses.

Kenyan Context

Kenyan Example:

Without Context:

```
"Help me write a loan application"
```

With Context:

```
"I am a 32-year-old mama mboga vendor in Nairobi's City Market. I've been
selling vegetables for 8 years. I make approximately Ksh 3,000-5,000 daily
profit. I have an M-Pesa transaction history but no formal bank account or
payslips. I need a Ksh 50,000 loan to expand my stall and buy a refrigerator.
I want to apply through M-Shwari. Help me write a compelling loan application
that highlights my reliability despite lacking traditional credit documents."
```

5.2.3 3. Instruction (Task)

Purpose: Clearly state what you want the AI to do. Be specific and actionable.

Instruction Type	Examples
Generate	"Write a 500-word blog post about..."
Summarize	"Condense this 10-page report into 5 key points..."
Analyze	"Identify the top 3 risks in this business plan..."
Compare	"Compare these two marketing strategies..."
Transform	"Convert this technical document into simple language..."
Create	"Design a 7-day content calendar for..."
Solve	"Calculate the break-even point for..."

Table 5.2: Instruction Types

5.2.4 4. Format/Constraints

Purpose: Specify how the output should be structured and any limitations.

Common Constraints:

- **Length:** "Keep it under 300 words" or "Write exactly 5 bullet points"

- **Format:** "Use markdown tables" or "Provide JSON format"
- **Tone:** "Professional and formal" or "Friendly and conversational"
- **Audience:** "Write for high school students" or "Write for CEOs"
- **Language:** "Use simple English" or "Include Swahili phrases"
- **Structure:** "Use headings and subheadings" or "Number each point"
- **Exclusions:** "Do not use technical jargon" or "Avoid mentioning competitors"

5.3 The Perfect Prompt Formula

The R-C-I-F Prompt Framework

Role + **C**ontext + **I**nstruction + **F**ormat = **Excellent Output**

Template:

```
Act as [ROLE].

[CONTEXT: Background information, situation, constraints]

Your task is to [INSTRUCTION: Specific action to perform]

Format the output as [FORMAT: Structure, length, style requirements]
```

5.3.1 Complete Example

Scenario: You need to create a social media post about a new product.

Complete Prompt:

```
Act as a creative social media manager specializing in African markets.

CONTEXT:
We are launching a new solar-powered phone charger called "SunCharge Pro"
targeting young professionals in Nairobi. Key features:
- Charges phone in 2 hours using sunlight
- Portable and lightweight (200g)
- Costs Ksh 3,500
- 2-year warranty
- Available at all Safaricom shops
- Eco-friendly, reduces carbon footprint

TASK:
Create an engaging Instagram post (caption + hashtags) that:
1. Highlights the product's benefits for busy professionals
2. Emphasizes environmental impact
3. Includes a call-to-action
```

```
4. Mentions the price and availability

FORMAT:
- Caption: 150-200 words
- Tone: Energetic, modern, slightly casual
- Include 10 relevant hashtags
- Use emojis strategically (3-5 max)
- Write in English with 1-2 Swahili phrases for local flavor
```

5.4 Iterative Prompting

5.4.1 The Iteration Process

Rarely do you get perfect results on the first try. Iterative prompting is the key to excellence.

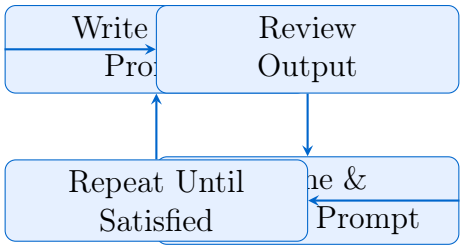


Figure 5.1: Iterative Prompting Cycle

5.4.2 Example of Iteration

Iteration 1:

```
"Write a business plan for a restaurant"
```

Output: Generic 2-page document with basic sections. Too vague.

Iteration 2:

```
"Write a business plan for a restaurant in Nairobi specializing in traditional
Kenyan cuisine. Include executive summary, market analysis, menu, financial
projections."
```

Output: Better, but still too general. Financial projections are unrealistic.

Iteration 3:

```
"Act as a restaurant consultant with 15 years experience in Nairobi. Write a
business plan for 'Nyama Choma Palace', a mid-range restaurant in Westlands
serving traditional Kenyan grilled meats.

Target market: Middle-class professionals aged 25-45
Seating capacity: 80 people
Startup capital: Ksh 2.5 million
Location: 150 sqm space, rent Ksh 80,000/month
```

```
Include:
1. Executive summary (200 words)
2. Market analysis with competitor comparison (Safari Inn, Carnivore)
3. Menu with pricing (20 items, average meal Ksh 800)
4. 12-month financial projections (realistic for Nairobi market)
5. Marketing strategy (digital + traditional)
6. Risk analysis and mitigation

Format as professional business document with headings, tables for financials."
```

Output: Comprehensive, realistic, actionable business plan.

5.4.3 Refinement Techniques

1. **Add Specificity:** Replace vague terms with exact numbers, names, dates
2. **Clarify Ambiguity:** If AI misunderstood, explicitly correct it
3. **Adjust Tone:** "Make it more formal" or "Sound more conversational"
4. **Change Length:** "Expand this section" or "Summarize in 100 words"
5. **Request Examples:** "Give me 3 concrete examples"
6. **Ask for Alternatives:** "Provide 3 different approaches"

5.5 Advanced Prompting Techniques

5.5.1 Few-Shot Prompting

Concept: Provide examples of what you want, and the AI will the pattern.

Example

Example: Customer Support Response Classification

```
Classify customer inquiries as URGENT, NORMAL, or LOW priority.

EXAMPLES:

Inquiry: "My account was hacked and money was stolen!"
Priority: URGENT
Reason: Security breach, financial loss

Inquiry: "When will my loan application be processed?"
Priority: NORMAL
Reason: Standard inquiry, no immediate risk

Inquiry: "Can you send me your brochure?"
```

```
Priority: LOW
Reason: Information request, no urgency

NOW CLASSIFY THIS:

Inquiry: "I've been waiting 3 weeks for my refund and still nothing!"
Priority: [AI will fill this]
Reason: [AI will fill this]
```

5.5.2 Chain-of-Thought Prompting

Concept: Ask the AI to think step-by-step before answering. This improves accuracy on complex problems.

```
Question: A farmer has 50 acres. She plants maize on 40% of the land, beans
on 30%, and keeps the rest fallow. Maize yields 2 tons per acre, beans yield
1.5 tons per acre. Maize sells for Ksh 50/kg, beans for Ksh 120/kg. Calculate
total revenue.

Solve this step-by-step:
1. Calculate acres for each crop
2. Calculate yield in tons for each crop
3. Convert tons to kilograms
4. Calculate revenue for each crop
5. Sum total revenue

Show all calculations clearly.
```

5.5.3 Role-Playing Multiple Perspectives

Concept: Ask AI to analyze from different viewpoints.

```
Analyze this business decision from three perspectives:

BUSINESS DECISION: Should a Nairobi-based e-commerce startup expand to rural
markets immediately or consolidate in urban areas first?

PERSPECTIVE 1: Optimistic Growth Investor
- Focus on: Market opportunity, first-mover advantage, scalability

PERSPECTIVE 2: Conservative CFO
- Focus on: Cash flow, risk management, unit economics

PERSPECTIVE 3: Operations Manager
- Focus on: Logistics, infrastructure, execution challenges

For each perspective, provide:
- Main argument (for or against)
- Top 3 supporting points
```

```
- Biggest concern
- Recommendation

Then synthesize: What is the balanced approach?
```

5.6 Common Prompting Mistakes

Mistake	Solution
Too Vague	Be specific about what you want. Instead of "Write something about marketing," say "Write a 500-word blog post about digital marketing strategies for Kenyan SMEs."
No Context	Provide background information. The AI doesn't know your situation unless you tell it.
Multiple Tasks	Ask one thing at a time. Don't say "Write a poem, then analyze it, then translate it." Break into separate prompts.
Assuming Knowledge	Don't assume the AI knows recent events or local context. Provide necessary information.
No Format Specification	Specify length, structure, tone. Otherwise you get unpredictable results.
Giving Up After One Try	Iterate! Refine your prompt based on the output.

Table 5.3: Common Prompting Mistakes and Solutions

5.7 Prompt Templates for Common Tasks

5.7.1 Template 1: Email Writing

```
Act as a [professional role, e.g., "senior marketing manager"].

CONTEXT:
```

```
I need to send an email to [recipient] about [topic].
Key points to include:
1. [Point 1]
2. [Point 2]
3. [Point 3]

The relationship is [formal/casual] and the tone should be [tone].

TASK:
Write a professional email that:
- Has a compelling subject line
- Opens with appropriate greeting
- Clearly states the purpose
- Includes all key points
- Has a clear call-to-action
- Closes professionally

FORMAT:
- Length: [e.g., 150-200 words]
- Tone: [e.g., professional but friendly]
- Include: [specific requirements]
```

5.7.2 Template 2: Data Analysis

```
Act as a data analyst specializing in [industry].

CONTEXT:
I have sales data for the past 6 months:
[Insert data or describe dataset]

TASK:
Analyze this data and provide:
1. Key trends and patterns
2. Best and worst performing [products/regions/months]
3. Year-over-year comparison (if applicable)
4. Three actionable insights
5. Potential risks or concerns

FORMAT:
- Use bullet points for clarity
- Include specific numbers and percentages
- Highlight important findings in bold
- Keep analysis under 500 words
- Suggest visualizations (charts/graphs) that would help
```

5.7.3 Template 3: Content Creation

```
Act as a content creator specializing in [topic] for [audience].
```

```
CONTEXT:
We are creating content about [topic] to [goal: educate/persuade/entertain].
Target audience: [describe audience demographics and interests]
Platform: [blog/Instagram/LinkedIn/YouTube]

TASK:
Create [type of content: blog post/social media caption/video script] that:
- Hooks the audience in the first sentence
- Covers these key points: [list points]
- Includes a compelling call-to-action
- Is optimized for [SEO/engagement/conversions]

FORMAT:
- Length: [specific word count or duration]
- Tone: [conversational/professional/inspirational]
- Style: [storytelling/data-driven/how-to]
- Include: [hashtags/emojis/links/images suggestions]
```

5.8 Practical Exercises

5.8.1 Exercise 1: Prompt Makeover

Take these weak prompts and transform them using the R-C-I-F framework:

1. "Help me with my business"
2. "Write a report"
3. "Explain machine learning"
4. "Create a marketing plan"
5. "Analyze this data"

For each, write:

- Original prompt
- Improved prompt with Role, Context, Instruction, Format
- Test both in ChatGPT and compare outputs

5.8.2 Exercise 2: Iterative Refinement

Task: Create a product description for a new smartphone.

1. Write initial prompt
2. Generate output
3. Identify 3 weaknesses in the output

4. Refine prompt to address weaknesses
5. Generate new output
6. Repeat 3 times
7. Document how the output improved with each iteration

5.8.3 Exercise 3: Build a Prompt Library

Create a personal prompt library with 10 templates for tasks you do regularly:

- Email responses
- Meeting notes summarization
- Social media posts
- Report writing
- Data analysis
- Code debugging
- Learning explanations
- Brainstorming sessions
- Customer responses
- Project planning

Save these in a document for repeated use.

5.9 Chapter Summary

- **Prompt Engineering is a Skill:** Like any skill, it improves with practice and deliberate learning.
- **Use the R-C-I-F Framework:** Role + Context + Instruction + Format = Excellent results.
- **Iterate Relentlessly:** First attempts are rarely perfect. Refine based on outputs.
- **Be Specific:** Vague prompts get vague results. Provide details, numbers, examples.
- **Advanced Techniques:** Few-shot prompting and chain-of-thought improve complex task performance.
- **Build Templates:** Create reusable prompt templates for common tasks to save time.
- **Avoid Common Mistakes:** Don't be vague, don't skip context, don't give up after one try.
- **Practice Daily:** The more you prompt, the better you become. Make it a habit.

Chapter 6

Applied Problem Solving and Group Work

Learning Objective

Chapter Learning Objectives:

- Apply a structured framework to identify AI opportunities
- Evaluate feasibility of AI solutions
- Work effectively in teams to design AI projects
- Present AI solutions clearly and persuasively
- Critique and improve peer proposals
- Plan implementation steps for AI projects

6.1 Introduction to Applied AI Problem Solving

6.1.1 From Theory to Practice

Throughout this course, you've learned about:

- AI concepts and technologies
- Tools and platforms
- Real-world applications in Kenya
- Prompt engineering techniques

Now it's time to **apply** this knowledge to solve real problems.

6.1.2 The AI Project Lifecycle

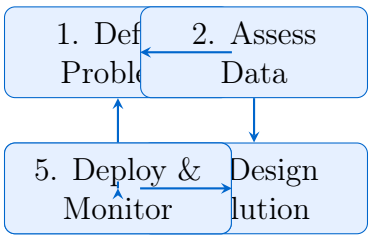


Figure 6.1: AI Project Lifecycle

6.2 Framework for Designing AI Solutions

6.2.1 Step 1: Problem Definition

Problem Definition Checklist

Ask These Questions:

1. **What is the specific problem?**
   - Describe the problem in one clear sentence
   - Avoid vague statements like "improve efficiency"
   - Be specific: "Reduce customer wait time from 30 minutes to 5 minutes"
2. **Who is affected?**
   - Identify primary stakeholders
   - Quantify impact: "500 customers daily affected"
   - Understand their pain points
3. **How often does it occur?**
   - Frequency: Daily, weekly, monthly?
   - Pattern: Predictable or random?
   - Trend: Getting better or worse?
4. **What is the economic/social impact?**
   - Financial cost: Ksh per month/year
   - Time cost: Hours lost
   - Opportunity cost: What could be done instead?
   - Social impact: Quality of life, equity, access
5. **How is it currently addressed?**
   - Current solution(s)
   - Why are they inadequate?
   - What are the limitations?

Example: Good vs. Bad Problem Definition

Bad Problem Definition	Good Problem Definition
"Farmers need help with their crops"	"Smallholder maize farmers in Uasin Gishu lose 30% of their yield to undetected crop diseases, resulting in Ksh 50,000 average annual loss per farmer"
"Improve customer service"	"Customer support team receives 200+ repetitive questions daily about order status, taking 15 minutes each to respond, totaling 50 hours/week of agent time"
"Use AI for business"	"Our e-commerce platform has 15% cart abandonment rate. We want to predict which users will abandon and send personalized discount offers to reduce abandonment to 10%"

Table 6.1: Problem Definition Examples

6.2.2 Step 2: Data Assessment

Kenyan Context

Critical Question: Do you have the data needed to solve this problem with AI?

Data Requirements Analysis

1. What data is needed?

- Type: Text, numbers, images, audio, video?
- Volume: How many examples?
- Quality: Accuracy, completeness, consistency?
- Labels: Do you have correct answers for training?

2. Is the data available?

☐ Already collected and digitized

☐ Exists but needs digitization

☐ Partially available

☐ Must be collected from scratch

3. Data Sources:

- Internal systems (databases, spreadsheets)
- Public datasets
- APIs (weather, maps, social media)
- Manual collection (surveys, sensors)
- Third-party providers

4. Data Quality Checklist:

☐ Complete (no missing values)

☐ Accurate (correct information)

☐ Consistent (standardized format)

☐ Relevant (related to problem)

☐ Timely (up-to-date)

☐ Unbiased (representative sample)

6.2.3 Step 3: Solution Design

AI Solution Design Template

Problem: [One-sentence description]

Proposed AI Solution:

- **Type of AI:** [Computer Vision / NLP / Predictive ML / Recommendation / Other]
- **How it works:** [2-3 sentences explaining the mechanism]
- **Input:** [What data goes in]
- **Output:** [What result comes out]

Technology Stack:

- **Tools:** [ChatGPT, Teachable Machine, AutoML, etc.]
- **Platform:** [Web app, mobile app, WhatsApp bot, etc.]
- **Integration:** [How it connects to existing systems]

User Journey:

1. User does [action]
2. System processes [data]
3. AI generates [output]
4. User receives [result]

Success Metrics:

- Primary: [Main KPI, e.g., "Reduce processing time by 50%"]
- Secondary: [Supporting metrics]
- Target: [Specific numbers]

6.2.4 Step 4: Feasibility Assessment

Dimension	Key Questions	Score (1-5)
Technical Feasibility	Can it be built with current tools and skills? Do we have/ can we get the data?	
Economic Feasibility	What is the cost (development, deployment, maintenance)? What is the ROI? Is it affordable?	
Operational Feasibility	Will it integrate with existing workflows? Do we have infrastructure (internet, electricity)?	
Cultural Feasibility	Will people use it? Is it culturally appropriate? Does it respect local norms?	
Legal/Ethical Feasibility	Does it comply with data protection laws? Is it fair and unbiased? Any privacy concerns?	

Table 6.2: Feasibility Assessment Matrix

Scoring Guide:

- 5 = Highly feasible (no major obstacles)
- 4 = Feasible (minor challenges, solvable)
- 3 = Moderately feasible (significant challenges)
- 2 = Difficult (major obstacles)
- 1 = Not feasible (insurmountable barriers)

Decision Rule:

- Total score 20-25: **Proceed** - Strong feasibility
- Total score 15-19: **Proceed with Caution** - Address challenges
- Total score <15: **Reconsider** - Major issues to resolve

6.3 Group Work Methodology

6.3.1 Team Roles

For effective group work, assign these roles:

Role	Responsibilities
Project Lead	Coordinates team, ensures deadlines met, facilitates meetings, final decision-maker
Researcher	Gathers data, researches similar solutions, identifies best practices, benchmarks
Technical Lead	Evaluates tools, designs technical architecture, assesses feasibility, plans implementation
User Experience (UX) Lead	Represents end-user perspective, designs user journey, ensures accessibility, tests usability
Presenter	Creates presentation materials, practices delivery, handles Q&A, ensures clarity

Table 6.3: Team Roles and Responsibilities

Note: In small teams (3-4 people), one person may hold multiple roles.

6.3.2 Collaboration Best Practices

1. **Establish Communication:**
   - Create WhatsApp/Telegram group
   - Schedule regular check-ins (e.g., twice weekly)
   - Use shared documents (Google Docs)
   - Set clear response time expectations
2. **Define Work Distribution:**
   - Assign tasks based on strengths
   - Set individual deadlines (before group deadline)
   - Use project management tools (Trello, Asana)
   - Document who does what
3. **Ensure Quality:**
   - Peer review each other's work
   - Consolidate early (don't wait until last minute)
   - Resolve conflicts through discussion
   - Escalate to instructor if needed
4. **Prepare Presentation:**

- Rehearse together at least twice
- Time your presentation (stick to limit)
- Anticipate questions
- Assign who answers what

6.4 Case Study Analysis

6.4.1 Case Study 1: AI for Maternal Health

Example

Scenario:

A rural county hospital in Kenya faces these challenges:

- High maternal mortality rate (300 deaths per 100,000 births vs. national average of 100)
- Only 2 midwives for 5,000 expectant mothers annually
- Many women deliver at home without skilled attendance
- Late detection of pregnancy complications
- Poor follow-up after delivery
- Limited health education in local languages

Your Task: Design an AI-powered solution to improve maternal health outcomes.
Deliverables:

1. Problem definition (use checklist)
2. Data assessment (what data exists/needed)
3. Proposed AI solution (type, how it works, tools)
4. Feasibility assessment (score each dimension)
5. Implementation plan (phases, timeline, resources)
6. Expected impact (quantified benefits)
7. Risks and mitigation strategies

6.4.2 Case Study 2: AI for Small Business Lending

Example

Scenario:

A microfinance institution (MFI) in Nairobi wants to expand lending to small businesses but faces:

- 40% loan default rate
- Time-consuming manual application processing (5-7 days)
- Lack of reliable credit information for informal businesses
- High operational costs (Ksh 2,000 per loan processed)
- Inability to scale (only 50 loans/month with current staff)
- Bias in lending decisions (favoring certain areas/ethnicities)

Your Task: Design an AI-powered credit scoring and loan processing system.

6.4.3 Case Study 3: AI for Education

Example

Scenario:

A primary school in a low-income area has:

- 80 students per class (1 teacher)
- 60% of students reading below grade level
- No individualized attention possible
- Teachers overwhelmed with grading
- Parents unable to help (many illiterate)
- Limited learning materials
- High dropout rate in upper primary

Your Task: Design an AI-powered personalized learning system.

6.5 Presentation Guidelines

6.5.1 Structure

Your group presentation (10-15 minutes) should include:

1. **Introduction (2 min)**
   - Team members and roles
   - Problem statement
   - Why it matters (impact)
2. **Problem Analysis (3 min)**
   - Who is affected
   - Current solutions and limitations
   - Data availability
3. **Proposed Solution (5 min)**
   - How the AI works
   - Technology stack
   - User journey
   - Feasibility assessment
4. **Implementation Plan (3 min)**
   - Phases and timeline
   - Resources needed
   - Success metrics
   - Risks and mitigation
5. **Q&A (2-5 min)**
   - Answer questions from audience
   - Defend your design choices
   - Acknowledge limitations

6.5.2 Presentation Tips

- **Visuals:** Use diagrams, flowcharts, mockups
- **Clarity:** Avoid jargon, explain technical terms
- **Engagement:** Tell a story, use examples
- **Practice:** Rehearse multiple times
- **Time Management:** Stick to allocated time
- **Teamwork:** All members should present

6.6 Capstone Project Preparation

6.6.1 Project Selection

For your Week 6 capstone project, choose from:

1. **Build an AI-powered workflow or automation**
   - Example: Automated customer inquiry response system
   - Tools: Zapier/Make + ChatGPT + Google Sheets
   - Deliverable: Working automation + documentation
2. **Train a simple ML model for a practical problem**
   - Example: Predict student performance, sales forecasting
   - Tools: Teachable Machine, AutoML, or Azure ML
   - Deliverable: Trained model + evaluation report
3. **Develop a real-world AI solution**
   - Example: Chatbot for local business, image classifier
   - Tools: Choice of platform based on need
   - Deliverable: Functional prototype + user guide

6.6.2 Project Development Steps

Follow this structured approach:

1. **Week 1-2: Problem Definition**
   - Identify problem
   - Research existing solutions
   - Define success criteria
   - Submit project proposal
2. **Week 3: Data Collection**
   - Gather dataset
   - Clean and prepare data
   - Split into train/test sets
   - Document data sources
3. **Week 4: Model Development**
   - Select tool/platform
   - Train initial model
   - Evaluate performance

- Iterate and improve

4. **Week 5: Testing and Refinement**
   - Test with real users/data
   - Gather feedback
   - Make improvements
   - Prepare documentation
5. **Week 6: Final Presentation**
   - Create presentation
   - Demonstrate solution
   - Submit final report
   - Reflect on learnings

6.7 Chapter Summary

- **Structured Approach:** Use the problem definition, data assessment, solution design, and feasibility framework for all AI projects.
- **Teamwork:** Assign clear roles, communicate regularly, and leverage each member's strengths.
- **Real-World Focus:** Always ground your solutions in actual user needs and local context.
- **Feasibility First:** Don't fall in love with technology—ensure the solution is practical, affordable, and sustainable.
- **Iterate:** Test early, gather feedback, and continuously improve.
- **Communicate:** Present your ideas clearly, use visuals, and tell a compelling story.
- **Ethics:** Always consider fairness, bias, privacy, and social impact.

6.8 Practical Exercises

6.8.1 Exercise 1: Problem Identification

In your community or workplace, identify 3 problems that could potentially be solved with AI. For each:

- Write a clear problem statement
- Quantify the impact (who, how many, how much)
- Describe current solutions and their limitations
- Assess data availability
- Propose a potential AI approach

6.8.2 Exercise 2: Feasibility Analysis

Take one of the problems from Exercise 1 and conduct a full feasibility assessment:

- Score each dimension (technical, economic, operational, cultural, legal)
- Justify your scores with evidence
- Identify top 3 challenges
- Propose mitigation strategies
- Make a go/no-go recommendation

6.8.3 Exercise 3: Group Project

Form a team of 3-4 students and:

- Select one case study from Section 4
- Assign roles
- Complete the full solution design
- Create a 10-minute presentation
- Present to the class
- Receive and incorporate feedback

Appendix A

Comprehensive Glossary of AI Terms

This glossary provides clear, jargon-free definitions of key terms used throughout the course. It is designed to be a quick reference guide for your studies and future projects.

mcblue	
Algorithm	A step-by-step set of instructions or mathematical rules that a computer follows to solve a specific problem or perform a task.
Artificial Intelligence (AI)	The broad field of computer science focused on creating systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, and problem-solving.
Artificial Neural Network (ANN)	A computing system inspired by the biological neural networks in the human brain. It consists of interconnected nodes (neurons) organized in layers that process information.
Automation	The use of technology to perform tasks with minimal human intervention. In AI, this often refers to automating repetitive cognitive or administrative tasks.
Bias (Algorithmic)	Systematic and repeatable errors in a computer system that create unfair outcomes, often stemming from biased or unrepresentative training data.
Big Data	Extremely large datasets that may be analyzed computationally to reveal patterns, trends, and associations relating to human behavior and interactions.
Chatbot	A software application designed to simulate human conversation through text or voice interactions, often used for customer service.
Classification	A type of supervised learning task where the model predicts a discrete label or category for a given input (e.g., "Spam" vs. "Not Spam").
Cloud Computing	The delivery of computing services (servers, storage, databases, AI tools) over the internet ("the cloud"), allowing users to access powerful tools without local hardware.

mcblue	
Computer Vision	A field of AI that enables computers to derive meaningful information from digital images, videos, and other visual inputs (e.g., facial recognition).
Convolutional Neural Network (CNN)	A specialized type of deep learning neural network primarily used for analyzing visual imagery and computer vision tasks.
Data	Raw, unorganized facts, numbers, text, or images that need to be processed. Data is the foundational "fuel" for all Machine Learning.
Data Cleaning	The process of detecting and correcting (or removing) corrupt, inaccurate, or irrelevant parts of a dataset to improve data quality.
Data Labeling	The process of adding meaningful tags or labels to raw data (like identifying objects in an image) so a machine learning model can learn from it.
Dataset	A structured collection of data, typically organized in tables (rows and columns) or files (like CSV), used for training and testing ML models.
Deep Learning (DL)	A specialized subset of machine learning based on multi-layered artificial neural networks capable of learning from vast amounts of unstructured data.
Edge Computing	Processing data near the source of generation (e.g., on a smartphone or IoT device) rather than sending it to a centralized cloud server, reducing latency.
Ethics (AI Ethics)	The moral principles and standards governing the development, deployment, and use of AI, focusing on fairness, transparency, privacy, and accountability.
Feature	An individual measurable property or characteristic of a phenomenon being observed. In a dataset, features are the input variables (columns).
Feature Engineering	The process of using domain knowledge to extract, select, and transform variables (features) from raw data to improve machine learning model performance.
Generative AI	A type of AI that can create new, original content (text, images, audio, code) based on patterns learned from its training data, rather than just analyzing existing data.
Hallucination	When an AI model (especially an LLM) generates information that is nonsensical, factually incorrect, or fabricated, but presents it confidently as truth.
Heuristic	A practical, problem-solving approach or "rule of thumb" that is not guaranteed to be optimal or perfect, but is sufficient for reaching an immediate goal.
Hyperparameter	A configuration variable set before the learning process begins (e.g., learning rate, number of clusters) that governs how the model is trained.

mcblue	
Inference	The process of using a trained machine learning model to make predictions or decisions on new, unseen data.
Internet of Things (IoT)	A network of physical objects ("things") embedded with sensors and connectivity, allowing them to collect and exchange data, often feeding AI models.
Label	The target variable or the "correct answer" in a supervised learning dataset (e.g., the word "Cat" for an image of a cat).
Large Language Model (LLM)	A highly advanced deep learning model trained on massive amounts of text data, capable of understanding, generating, and translating human language (e.g., ChatGPT).
Machine Learning (ML)	A subset of AI where systems learn patterns from data and improve their performance on a specific task over time without being explicitly programmed.
Model	The output artifact generated after an ML algorithm has been trained on a dataset. It represents the learned patterns and is used to make predictions.
Natural Language Processing (NLP)	A branch of AI focused on enabling computers to understand, interpret, manipulate, and respond to human language (text or voice).
Overfitting	When a machine learning model learns the training data too well, including its noise and outliers, resulting in poor performance on new, unseen data.
Predictive Analytics	The use of data, statistical algorithms, and ML techniques to identify the likelihood of future outcomes based on historical data.
Prompt	The input text, question, or instruction provided by a user to guide a Generative AI model in generating a specific output.
Prompt Engineering	The practice of carefully designing, refining, and structuring prompts to elicit the most accurate, relevant, and high-quality responses from an AI model.
Regression	A type of supervised learning task where the model predicts a continuous numerical value (e.g., predicting house prices or future sales).
Reinforcement Learning	A type of ML where an agent learns to make decisions by performing actions in an environment and receiving rewards or penalties (feedback) for those actions.
Robotic Process Automation (RPA)	Technology that uses software "robots" to automate highly repetitive, rule-based digital tasks, often combined with AI for intelligent automation.
Supervised Learning	A machine learning approach where the model is trained on a labeled dataset, meaning the input data is paired with the correct output (the answer key).

mcblue	
Test Data (Testing Set)	A subset of data used to evaluate the final performance of a trained machine learning model. The model never sees this data during training.
Training Data (Training Set)	The subset of data used to teach the machine learning model. The algorithm analyzes this data to learn patterns and build the model.
Underfitting	When a machine learning model is too simple to capture the underlying patterns in the training data, resulting in poor performance on both training and new data.
Unsupervised Learning	A machine learning approach where the model is trained on unlabeled data and must find hidden structures, patterns, or groupings on its own.
Validation Data (Validation Set)	A subset of data used during the training process to tune hyperparameters and prevent overfitting, acting as a bridge between training and testing.
Virtual Assistant	An AI-powered software agent that can perform tasks or services for an individual, often utilizing NLP and voice recognition (e.g., Siri, Alexa).
Weights and Biases	Internal parameters within a neural network that are adjusted during training. They determine the strength and nature of the connections between neurons.

Appendix B

Recommended Reading and Resources

To deepen your understanding and stay updated with the rapidly evolving field of AI, we highly recommend the following resources.

B.1 Books

B.1.1 Beginner Friendly (No Coding Required)

- **"Artificial Intelligence: A Guide for Thinking Humans"** by Melanie Mitchell. *Excellent for conceptual understanding without heavy math.*
- **"The Hundred-Page Machine Learning Book"** by Andriy Burkov. *A concise, brilliant overview of ML concepts.*
- **"AI Superpowers: Competition, Cooperation, and the New World Order"** by Kai-Fu Lee. *Great for understanding the global and economic impact of AI.*

B.1.2 Intermediate (Light Coding/Conceptual Depth)

- **"Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow"** by Aurélien Géron. *The gold standard for practical ML implementation.*
- **"Deep Learning with Python"** by François Chollet. *Written by the creator of Keras, highly intuitive and practical.*

B.2 Online Courses and Platforms

Platform	Recommended Course	Cost
Elements of AI	Elements of AI (University of Helsinki)	Free
Coursera	AI For Everyone (Andrew Ng)	Free (Audit) / Paid
Kaggle	Intro to Machine Learning / Pandas	Free
Google Cloud	Google Cloud Skills Boost (AI Track)	Free / Paid
Microsoft	AI for Beginners (GitHub Curriculum)	Free

Table B.1: Top Online AI Learning Platforms

B.3 Websites, Blogs, and Newsletters

- **Towards Data Science (Medium):** Excellent articles on ML concepts, tutorials, and case studies.
- **Machine Learning Mastery:** Great practical tutorials and code examples.
- **The Batch (by DeepLearning.AI):** Andrew Ng's weekly newsletter summarizing the most important AI news.
- **TechCabal & Disrupt Africa:** Essential reading for tracking AI and tech startup developments across the African continent.

B.4 Communities and Local Context

Kenyan Context

Engage with the Local Ecosystem: Connecting with local peers is crucial for understanding how AI applies to the Kenyan context.

- **AI Kenya Community:** Look for local meetups, Twitter/X spaces, and LinkedIn groups discussing AI in Kenya.
- **Nairobi AI & Data Science Meetups:** Check platforms like Meetup.com or Eventbrite for local networking events.
- **iHub & Nairobi Garage:** Frequently host tech talks, hackathons, and workshops related to AI and innovation.

Appendix C

Assignment Templates and Rubrics

This appendix provides standardized templates and grading rubrics for all major assessments in the course.

C.1 Week 1 Assignment: AI in Your Daily Life

Submission Template: Week 1

Student Name: _____ **Date:** _____

Tool 1:

- **Name of Tool/Feature:**
- **Specific Task it Performs:**
- **Hypothesis on Data Used:**

Tool 2:

- **Name of Tool/Feature:**
- **Specific Task it Performs:**
- **Hypothesis on Data Used:**

Tool 3:

- **Name of Tool/Feature:**
- **Specific Task it Performs:**
- **Hypothesis on Data Used:**

Criteria	Marks	Description
Identification of 3 distinct AI tools	3	Tools are genuinely AI-driven, not just basic software.
Clear description of task performed	3	Accurately explains what the AI achieves for the user.
Logical hypothesis on data used	3	Reasonable guess on the input data required for the AI to function.
Formatting and Clarity	1	Professional layout, clear language, free of major errors.
Total	**10**	

Table C.1: Week 1 Assignment Rubric

C.2 Week 2 Assignment: Data Cleaning Documentation

Submission Template: Week 2

Dataset Name: _____ **Original Rows:** _____

Step 1: Handling Missing Values

- **Columns with missing data:**
- **Action taken (Drop/Fill):**
- **Justification:**

Step 2: Removing Duplicates

- **Number of duplicates found:**
- **Action taken:**

Step 3: Data Formatting & Validation

- **Inconsistencies fixed (e.g., text casing, date formats):**
- **Outliers removed or adjusted:**

Final Dataset Stats: _____ **Rows,** _____ **Columns.**

C.3 Week 6 Capstone Project Rubric

Criteria	Marks	Description
Problem Definition & Relevance	15	Clear, specific problem statement. High relevance to local/Kenyan context or business needs.
Data Preparation & Quality	15	Dataset is appropriately cleaned, formatted, and split. Clear understanding of features and labels.
Model Selection & Training	20	Appropriate tool/algorithm chosen. Correct training process followed. Hyperparameters considered.
Evaluation & Interpretation	20	Model evaluated using correct metrics (Accuracy, Precision, etc.). Results are interpreted in plain language.
Functionality & Automation	15	The workflow, automation, or model functions as intended without critical errors.
Presentation & Documentation	15	Clear, engaging presentation. Professional documentation. Ability to answer Q&A confidently.
Total	**100**	

Table C.2: Capstone Project Grading Rubric

Appendix D

Technical Setup Guide

This guide provides step-by-step instructions to ensure your technical environment is fully prepared for the hands-on sessions.

D.1 System Requirements Verification

Before proceeding, ensure your machine meets the baseline requirements:

Important

Minimum Specs: Windows 10+, macOS 10.15+, or Linux | 256 GB SSD | 4 GB RAM (8 GB recommended) | 5 Mbps Internet.

D.2 Account Setup: Google Ecosystem

Many of our tools (Colab, Sheets, Teachable Machine, AutoML) require a Google Account.

1. Go to `accounts.google.com` and click **Create account** (Choose "For my personal use").
2. Follow the prompts to set up your Gmail address and password.
3. **Verify your phone number** if prompted (crucial for accessing certain cloud services).
4. **Test Access:** Open a new tab and go to `drive.google.com` to ensure you are logged in.

D.3 Account Setup: Generative AI Tools

You will need access to at least one major Generative AI platform. We recommend setting up both for comparison.

D.3.1 ChatGPT (OpenAI)

1. Navigate to `chat.openai.com`.
2. Click **Sign up** and use your Google Account or email address.
3. Verify your email and set a secure password.
4. **Tip:** Bookmark the URL for quick access during practical sessions.

D.3.2 Google Gemini

1. Navigate to `gemini.google.com`.
2. Since you are already logged into your Google Account, simply click **Get Started** and accept the terms of service.
3. You are now ready to use Gemini.

D.4 Tool Setup: Google Teachable Machine

Teachable Machine runs entirely in your browser and requires no installation.

Example

Setup Steps:

1. Go to `teachablemachine.withgoogle.com`.
2. Click **Get Started**.
3. Select **Standard image model** (recommended for beginners).
4. **Crucial Step:** Your browser will ask for permission to use your **Camera** and **Microphone**. You **MUST** click **Allow**. If you block this, the tool will not work.

D.5 Tool Setup: Google Colab

Google Colab provides a free, cloud-based Python environment (Jupyter Notebook) for when we touch upon light coding.

1. Go to `colab.research.google.com`.
2. Click **File** → **New notebook**.
3. A new notebook will open. In the first cell, type: `print("Hello Mediacrest!")`
4. Press **Shift** + **Enter** to run the cell. If you see the output, your setup is successful.
5. **Tip:** Colab requires a stable internet connection. If you disconnect, you may need to reconnect to the runtime.

D.6 Troubleshooting Common Issues

Issue	Solution
Camera not working in Teachable Machine	Check browser settings. Ensure Chrome/Edge has permission to access the camera. Close other apps using the camera (like Zoom).
Colab disconnects frequently	This usually happens due to inactivity or poor internet. Keep the tab active. If it drops, click "Reconnect" in the top right.
ChatGPT says "Capacity Reached"	The free tier gets busy during peak hours. Try using Google Gemini or Microsoft Copilot as an alternative.
Browser running slowly	Clear your browser cache, disable unnecessary extensions, or switch to a different supported browser (e.g., from Chrome to Edge).

Table D.1: Common Technical Troubleshooting

Kenyan Context

Internet Connectivity Tip: If you are using mobile broadband, ensure you have sufficient data bundles before starting heavy cloud-based tools like Colab or AutoML. If your connection drops, most tools (like Colab and Teachable Machine) will save your progress locally or in your Google Drive, allowing you to resume once reconnected.

Bibliography

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[3] Burkov, A. (2019). *The Hundred-Page Machine Learning Book*. Andriy Burkov.

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[5] Chollet, F. (2021). *Deep Learning with Python* (2nd ed.). Manning Publications.